

CHAPTER 1

Concept of Basic Electrical and Electronics Engineering

Complete Textbook for NEC License Examination Preparation

Electronics, Communication and Information Engineering (AEiE)

Nepal Engineering Council

Registration Examination

Code: AExE01

Master Roadmap

Basic Concept (Part 1)

Basic Concept (Part 2)

Network Theorems

AC Fundamentals

Semiconductor Devices

Signal Generators

Amplifiers

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CHAPTER 1 — MASTER ROADMAP

Concept of Basic Electrical and Electronics Engineering (AExE01)

1. Complete Hierarchical Structure

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- └─ 1.1 Basic Concept (AExE0101)
 - └─ 1.1.1 Electric Charge (prerequisite)
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 - └─ 1.2.7 R-L-C Circuits (DC & AC)
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 - └─ Applications
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 - └─ Transistor Action
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 - └─ Operating Regions
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 - └─ 1.6.9 Tuned Amplifiers
 - └─ 1.6.10 Operational Amplifiers (Op-Amps)
 - └─ Ideal vs Practical Op-Amp
 - └─ Open-Loop & Closed-Loop Gain
 - └─ Virtual Short & Virtual Ground
 - └─ Inverting Amplifier
 - └─ Non-Inverting Amplifier
 - └─ Voltage Follower
 - └─ Summing Amplifier
 - └─ Difference Amplifier
 - └─ Integrator & Differentiator
 - └─ Comparator

2. Dependency Relationships

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graph TD
    A["1.1 Basic Concept<br/>(Charge, V, I, R, P, E)"] --> B["1.1 Circuit Types<br/>(Series, Parallel, Star-Delta)"]
    A --> C["1.1 Kirchhoff's Laws<br/>(KCL, KVL)"]
    B --> C
    C --> D["1.2 Network Theorems<br/>(Superposition, Thevenin, Norton, MPT)"]
    A --> E["1.3 AC Fundamentals<br/>(Generation, Waveforms, Values)"]
    E --> F["1.2 R-L, R-C, R-L-C Circuits"]
    E --> G["1.2 Resonance<br/>(Series & Parallel)"]
    F --> G
    E --> H["1.2 Active, Reactive,<br/>Apparent Power"]
    E --> I["1.3 Three-Phase System"]
    A --> J["1.4 Semiconductor<br/>Fundamentals"]
    J --> K["1.4 Diode"]
    K --> L["1.4 BJT"]
    L --> M["1.4 MOSFET"]
    M --> N["1.4 CMOS"]
    L --> O["1.5 Oscillator<br/>Principles"]
    D --> O
    O --> P["1.5 RC Oscillators"]
    O --> Q["1.5 LC Oscillators"]
    O --> R["1.5 Crystal Oscillators"]
    L --> S["1.6 Amplifier<br/>Classes A/B/AB"]
    S --> T["1.6 Push-Pull &<br/>Tuned Amplifiers"]
    D --> U["1.6 Op-Amps"]
    S --> U
```

Key dependency insight: Section 1.2 spans both DC analysis (network theorems) and AC analysis (R-L-C, resonance, power). AC fundamentals from §1.3 are needed to fully understand parts of §1.2. The recommended approach is to study §1.1 → §1.3 (AC basics only) → §1.2 (full section), or study §1.2 network theorems first (DC), then §1.3, then return for AC parts of §1.2. We will follow the syllabus order but introduce AC concepts as needed.

3. Recommended Learning Order

ORDER	TOPIC	RATIONALE
1	§1.1 — Charge, Voltage, Current, Resistance, Ohm's Law	Foundation for everything
2	§1.1 — Power, Energy	Builds on V, I, R
3	§1.1 — Conductors, Insulators	Material classification
4	§1.1 — Series & Parallel Circuits	First circuit analysis
5	§1.1 — Star-Delta, Delta-Star Conversion	Circuit transformation
6	§1.1 — Kirchhoff's Laws (KCL, KVL)	Core analysis tool
7	§1.1 — Circuit Classifications	Conceptual framework
8	§1.2 — Superposition Theorem	First network theorem (uses KVL/KCL)
9	§1.2 — Thevenin's Theorem	Most important theorem
10	§1.2 — Norton's Theorem	Dual of Thevenin
11	§1.2 — Maximum Power Transfer	Builds on Thevenin
12	§1.3 — AC Generation, Equations, Waveforms	AC foundation
13	§1.3 — Peak, Average, RMS Values	AC measurement
14	§1.3 — Phasor Representation	AC analysis tool
15	§1.2 — R-L, R-C, R-L-C Circuits (AC context)	AC circuit analysis
16	§1.2 — Impedance, Reactance	AC circuit parameters
17	§1.2 — Series Resonance	AC special case
18	§1.2 — Parallel Resonance	AC special case
19	§1.2 — Active, Reactive, Apparent Power	AC power
20	§1.3 — Three-Phase System	AC power distribution
21	§1.4 — Semiconductor Fundamentals	Electronics foundation

ORDER	TOPIC	RATIONALE
22	§1.4 — Diode	First device
23	§1.4 — BJT	Core transistor
24	§1.4 — MOSFET	Field-effect transistor
25	§1.4 — CMOS	Digital electronics bridge
26	§1.5 — Oscillator Principles	Feedback theory
27	§1.5 — RC, LC, Crystal Oscillators	Specific circuits
28	§1.5 — Waveform Generators	Signal shaping
29	§1.6 — Amplifier Fundamentals	Gain concepts
30	§1.6 — Class A, B, AB Stages	Output classifications
31	§1.6 — Push-Pull, Tuned Amplifiers	Power stages
32	§1.6 — Op-Amps	Integrated amplifiers

4. Priority Classification

HIGH PRIORITY — Frequently tested, fundamental, high-value

TOPIC	WHY HIGH PRIORITY
Ohm's Law	Foundation of all circuit analysis
Kirchhoff's Laws (KCL, KVL)	Required for network theorem problems
Series & Parallel Circuits	Basic circuit analysis
Star-Delta / Delta-Star Conversion	Frequently examined transformation
Thevenin's Theorem	Most important network theorem
Norton's Theorem	Frequently tested alongside Thevenin
Maximum Power Transfer	Classic exam problem
RMS, Average, Peak Values	Essential AC calculations
R-L-C Circuits	Core AC analysis
Series & Parallel Resonance	High-value exam topic
Active, Reactive, Apparent Power	Power calculations
Three-Phase Power Relationships	Standard exam topic
Diode Characteristics	Electronics fundamental
BJT Configurations & Biasing	Most tested transistor topic
BJT Current Relationships (α , β)	Numerical problems
Barkhausen Criterion	Oscillator fundamental
Op-Amp Circuits (Inverting, Non-Inverting)	High-frequency exam topic
Class A, B, AB Comparison	Conceptual and numerical

MEDIUM PRIORITY — Important but less frequently the main focus

TOPIC	WHY MEDIUM PRIORITY
Power and Energy	Often tested indirectly
Superposition Theorem	Important but less frequent than Thevenin
R-L, R-C Circuits (individually)	Usually tested as part of R-L-C
Quality Factor, Bandwidth	Resonance sub-topics
Phasor Representation	Tool for AC analysis
Small-Signal Model of BJT	Tested in amplifier context
MOSFET Regions of Operation	Less frequent than BJT but important
CMOS Inverter	Logic bridge topic
RC and LC Oscillator Circuits	Frequency calculations
Crystal Oscillator	Conceptual understanding
Push-Pull Amplifier	Power amplifier topic
Tuned Amplifier	Selective amplification
Summing & Difference Amplifier	Op-amp applications

● LOW PRIORITY — Still within syllabus, less central to exam

TOPIC	WHY LOW PRIORITY (BUT STILL COVERED)
Electric Charge (as standalone)	Usually implicit in other topics
Conducting vs Insulating Materials (detail)	Mostly conceptual/qualitative
Linear vs Non-linear Classification	Definition-level testing
Bilateral vs Unilateral Classification	Definition-level testing
Active vs Passive Classification	Definition-level testing
Form Factor, Crest Factor	Rarely tested directly
Large-Signal Model of BJT	Less frequent than small-signal
Waveform Generators (detail)	Typically conceptual
Power BJTs (thermal details)	Specialized sub-topic
Integrator, Differentiator (Op-Amp)	Less frequent than basic configs

Low priority ≠ skip. Every item will be covered. Priority guides study time allocation and revision order.

5. Topic Characterization

Numerical-Heavy Topics 1 2 3 4

- Ohm's Law calculations
- Series/Parallel resistance, voltage divider, current divider
- Star-Delta / Delta-Star conversion
- Kirchhoff's Laws (mesh & nodal analysis)
- Superposition, Thevenin, Norton numerical procedures

- Maximum Power Transfer calculations
- RMS, Average, Peak value calculations
- Impedance calculations (R-L, R-C, R-L-C)
- Resonant frequency, Q-factor, bandwidth
- Active/Reactive/Apparent power, power factor
- Three-phase voltage, current, power
- Diode circuit calculations
- BJT biasing point (Q-point) calculations
- Transistor parameter calculations (α , β , I_C , I_B , I_E)
- MOSFET drain current calculations
- Oscillator frequency calculations
- Amplifier gain and efficiency calculations
- Op-amp gain calculations

Formula-Heavy Topics

- Ohm's Law variants
- Power formulas ($P = VI$, $P = I^2R$, $P = V^2/R$)
- Star-Delta transformation formulas
- AC waveform equations ($v = V_m \sin(\omega t + \phi)$)
- Impedance formulas ($Z = R + jX$)
- Resonance formulas (f_o , Q , BW)
- Power formulas (P , Q , S , pf)
- Three-phase relationships ($V_L = \sqrt{3} V_{ph}$)
- BJT relationships ($I_E = I_C + I_B$, $\beta = I_C/I_B$)
- MOSFET current equations
- Oscillator frequency formulas
- Op-amp gain formulas

Diagram-Heavy Topics

- Series and parallel circuits
- Star and Delta connections
- Kirchhoff's Laws (circuit loops and nodes)
- Thevenin/Norton equivalent circuits
- AC waveforms (sinusoidal, phasor diagrams)

- R-L, R-C, R-L-C circuit diagrams and phasor diagrams
- Resonance curves (impedance vs frequency)
- Power triangle
- Three-phase connection diagrams and phasor diagrams
- PN junction and depletion region
- Diode V-I characteristics
- BJT construction and circuit configurations (CE, CB, CC)
- BJT characteristic curves
- MOSFET construction and characteristics
- CMOS inverter circuit and transfer characteristic
- Oscillator circuit diagrams (Wien Bridge, Phase-Shift, Hartley, Colpitts, Crystal)
- Amplifier circuit diagrams (Class A, B, AB, Push-Pull)
- Op-amp circuit configurations

Concept-Heavy Topics

- Conductor vs Insulator vs Semiconductor
- Linear vs Non-linear, Bilateral vs Unilateral, Active vs Passive
- Physical meaning of resonance
- Physical meaning of power factor
- Semiconductor doping (N-type, P-type)
- PN junction operation
- Transistor action (carrier injection & collection)
- CMOS complementary operation
- Positive feedback and oscillation condition
- Amplifier class comparison

Derivation-Heavy Topics

- Star-Delta / Delta-Star transformation formulas
- Average and RMS values of sinusoidal waveforms
- Three-phase line-phase relationships
- Resonant frequency from impedance condition
- Quality factor derivation
- Maximum power transfer condition
- Thevenin/Norton equivalence

- Oscillator frequency formulas (Wien Bridge, Phase-Shift, Hartley, Colpitts)
 - Op-amp gain formulas (from ideal op-amp assumptions)
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6. Commonly Confused Topics ⚡

CONFUSION	WHY IT'S CONFUSING
Voltage vs EMF	Both measured in volts; EMF is source, voltage is across element
Power vs Energy	$P = dE/dt$; power is rate, energy is total
Average vs RMS value	Different integrals; RMS is for power calculations
Resistance vs Reactance vs Impedance	R is real, X is imaginary, Z is complex
Inductive vs Capacitive reactance sign	$X_L = +\omega L$, $X_C = -1/(\omega C)$ in convention
Series vs Parallel resonance behavior	Series: min Z, max I; Parallel: max Z, min I
Active vs Reactive power	P does work, Q oscillates
Line vs Phase quantities (3 ϕ)	Depends on Star or Delta connection
Star-Delta vs Delta-Star direction	Formulas are different — not interchangeable
Thevenin vs Norton source deactivation	Voltage source $\rightarrow$ short; Current source $\rightarrow$ open
BJT Cutoff vs Saturation	Cutoff: both junctions reverse biased; Saturation: both forward
α vs β	$\alpha = I_C/I_E (< 1)$; $\beta = I_C/I_B (>> 1)$
CE vs CB vs CC	Which terminal is common to input and output
Enhancement vs Depletion MOSFET	Enhancement: normally OFF; Depletion: normally ON
Positive vs Negative feedback	Positive $\rightarrow$ oscillation; Negative $\rightarrow$ stable amplification
Class A vs B vs AB efficiency	A: 25% (max), B: 78.5% (max), AB: between

7. Expected Difficulty Rating

SECTION	CONCEPTUAL	MATHEMATICAL	OVERALL	NOTES
§1.1 Basic Concept	★★	★★	★★	Foundation — must be rock-solid
§1.2 Network Theorems	★★★★	★★★★★	★★★★★	Procedural and calculation-intensive
§1.2 Resonance & Power	★★★★★	★★★★★	★★★★★	Requires AC concepts from §1.3
§1.3 AC Fundamentals	★★★★	★★★★	★★★★	Integration for RMS/Average values
§1.3 Three-Phase	★★★★	★★★★	★★★★	Conceptually rich, formula-based calculations
§1.4 Diode	★★	★★	★★	Relatively straightforward
§1.4 BJT	★★★★★	★★★★★	★★★★★	Multiple configurations, biasing, models
§1.4 MOSFET & CMOS	★★★★	★★★★	★★★★	Conceptual understanding critical
§1.5 Oscillators	★★★★	★★★★	★★★★	Feedback + frequency derivations
§1.6 Amplifier Classes	★★★★	★★★★	★★★★	Efficiency calculations, comparisons
§1.6 Op-Amps	★★★★	★★★★	★★★★	Multiple configurations to master

(★ = Easy, ★★★★★ = Very Difficult)

8. Recommended Revision Order (After Full Study)

PRIORITY	REVISE FIRST	REASON
1st	Thevenin & Norton Theorems	Most exam-critical theorem
2nd	Kirchhoff's Laws + Ohm's Law	Foundation for all problems
3rd	Resonance (Series & Parallel)	High-value, formula-rich
4th	BJT Configurations & Biasing	Most tested electronics topic
5th	Op-Amp Circuits	Frequently tested, formula-based
6th	Three-Phase System	Standard power engineering questions
7th	Active/Reactive/Apparent Power	Power triangle mastery
8th	RMS, Average, Peak Values	AC value calculations
9th	Star-Delta Conversion	Transformation formulas
10th	Amplifier Classes (A, B, AB)	Comparison and efficiency
11th	Maximum Power Transfer	Builds on Thevenin
12th	Superposition Theorem	Multi-source problems
13th	Oscillator Circuits & Frequencies	Frequency formulas
14th	Diode Characteristics	Forward/Reverse bias
15th	MOSFET & CMOS	Working principle
16th	R-L, R-C, R-L-C Circuits	Impedance calculations
17th	Circuit Classifications	Quick recall
18th	Waveform Generators	Conceptual
19th	Comparison Tables	Final review

9. Cross-Connection Map

The chapter forms a connected knowledge graph:

FOUNDATION	ENABLES
Ohm's Law + KVL/KCL	→ Network Theorems → Thevenin/Norton
Thevenin's Theorem	→ Maximum Power Transfer
AC Waveform Equations	→ RMS/Average Values → Impedance → Resonance → Power
Semiconductor Physics	→ Diode → BJT → MOSFET → CMOS
BJT + Feedback Theory	→ Oscillators
BJT + Biasing	→ Amplifier Classes
Network Theorems + Feedback	→ Op-Amp Analysis

Study Strategy: Master §1.1 completely before moving forward. The entire chapter builds on these fundamentals. If KVL/KCL and series/parallel analysis are shaky, every subsequent topic will be harder than it needs to be.

Ready to Begin

Upon your approval, I will begin with:

§1.1 Basic Concept — Part 1: Electric Charge → Voltage → Current → Resistance → Ohm's Law → Power → Energy

This will be a comprehensive, textbook-quality treatment with definitions, derivations, diagrams, worked examples, and numerical problems.

CHAPTER 1 — CONCEPT OF BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Section 1.1 — Basic Concept (AExE0101)

PART 1: Electric Charge, Voltage, Current, Resistance, and Ohm's Law

Prerequisite Note

This section builds foundational concepts from the ground up. Even if you studied these in your BE program, review them carefully. NEC exam questions often test subtle understanding of fundamentals — not just formula recall.

1.1.1 ELECTRIC CHARGE

Introduction

Electric charge is the most fundamental quantity in electrical engineering. Every electrical phenomenon — from a simple flashlight circuit to a complex communication system — originates from the existence and movement of electric charges. Without understanding charge, nothing else in this chapter makes sense.

Why Is It Important?

An engineer needs the concept of charge because:

- **Voltage** is defined as energy per unit charge
- **Current** is defined as the rate of flow of charge
- **Capacitance** stores charge
- **Coulomb's law** governs force between charges
- All electromagnetic phenomena arise from charges (stationary or moving)

Basic Concept

All matter is composed of atoms. Each atom contains:

PARTICLE	LOCATION	CHARGE
Proton	Nucleus	Positive (+)
Neutron	Nucleus	Neutral (0)
Electron	Orbits around nucleus	Negative (–)

In a neutral atom, the number of protons equals the number of electrons. When an atom **loses** electrons, it becomes positively charged (cation). When it **gains** electrons, it becomes negatively charged (anion).

The fundamental unit of charge is the charge of one electron (or proton).

Definition

Electric charge is a fundamental physical property of matter that causes it to experience a force when placed in an electromagnetic field. It is a scalar quantity measured in **coulombs (C)**.

Physical Meaning

Charge is not something you can see or touch directly. What you observe are its **effects**:

- Two like charges repel each other
- Two unlike charges attract each other
- Moving charges create magnetic fields
- Accelerating charges radiate electromagnetic waves

Think of charge as a "label" on certain particles that determines how they interact with electric and magnetic fields.

Mathematical Formulation

The charge of a single electron:

$$e = 1.602 \times 10^{-19} \text{ C}$$

If a body has lost or gained n electrons, its net charge is:

$$Q = n \times e$$

SYMBOL	MEANING	SI UNIT
Q	Electric charge	Coulomb (C)
n	Number of electrons transferred	Dimensionless
e	Charge of one electron	$1.602 \times 10^{-19} \text{ C}$

Important Relationships

$$1\text{ C} = 11.602 \times 10^{-19} \approx 6.242 \times 10^{18} \text{ electrons}$$

One coulomb is an **enormous** amount of charge. In practical circuits, charges of microcoulombs (μC) or even picocoulombs (pC) are common in electronic devices.

Properties of Electric Charge

1. **Quantization**: Charge always exists in integer multiples of e . You cannot have $0.5e$.
2. **Conservation**: Charge can neither be created nor destroyed. The total charge in an isolated system remains constant.
3. **Additive**: Total charge = algebraic sum of individual charges.

Engineering Interpretation

In circuit analysis, we rarely think about individual electrons. Instead, we work with **current** (rate of charge flow) and **voltage** (energy per unit charge). However, understanding that current is fundamentally moving charges helps you understand:

- Why conductors conduct (free electrons)
- Why insulators don't (bound electrons)
- Why semiconductors are special (controllable conductivity)



Engineering Intuition

Think of charge like water. You don't usually measure individual water molecules — you measure flow rate (current) and pressure (voltage). But knowing that water is made of molecules helps you understand why certain pipes (conductors) let water through and others (insulators) don't.

1.1.2 ELECTRIC VOLTAGE (POTENTIAL DIFFERENCE)

Introduction

Voltage is arguably the most important concept in electrical engineering. It is the "driving force" that pushes electric charges through a circuit. Without voltage, there is no current, no power, and no useful work.

Why Is It Important?

- Voltage determines whether current will flow
- Voltage ratings define the operating limits of every electrical device
- Voltage measurement is the most common electrical measurement
- Understanding voltage is essential for applying Kirchhoff's Voltage Law (KVL)
- Power calculation requires voltage: $P=VI$

Basic Concept — Building From Energy

To move a charge against an electric field, you must do **work** on it (spend energy). This is analogous to lifting a mass against gravity — you expend energy, and the mass gains potential energy.

Similarly, when work is done to move a charge from one point to another in an electric field, the charge gains **electrical potential energy**.

Voltage (potential difference) is defined as the work done per unit charge in moving a charge between two points.

Definition

Voltage (or **potential difference**) between two points A and B is the work done per unit positive charge in moving a test charge from B to A.

$$V_{AB} = W/Q$$

Or equivalently:

Voltage is the energy transferred per unit charge.

Physical Meaning

Voltage represents an **energy difference** between two points. It tells you:

- How much energy each coulomb of charge gains (at a source) or loses (across a load) as it moves between those two points
- The "electrical pressure" driving charges through a circuit

Higher Potential (+) Lower Potential (-)

A ————— B

$$V_{AB} = V_A - V_B$$

Conventional current flows from A to B (high to low potential)

Electrons actually move from B to A (low to high potential)

Voltage is always measured BETWEEN two points. Saying "the voltage at point A" actually means "the voltage at point A with respect to some reference point (usually ground)."

Mathematical Formulation

$$V = W/Q$$

SYMBOL	MEANING	SI UNIT
V	Voltage (potential difference)	Volt (V)
W	Work done / Energy transferred	Joule (J)
Q	Charge	Coulomb (C)

Dimensional Check

$$[V] = [W]/[Q] = \text{J/C} = \text{V}$$

1 Volt = 1 Joule per Coulomb

This means: If 1 joule of energy is needed to move 1 coulomb of charge between two points, the potential difference between those points is 1 volt.

EMF vs Voltage Drop

Two related but distinct concepts:

QUANTITY	SYMBOL	MEANING
EMF (Electromotive Force)	$\mathcal{E}$ or E	Energy supplied per unit charge by a source (battery, generator)
Voltage Drop	V	Energy consumed per unit charge by a circuit element (resistor, load)

- A **battery** provides EMF — it converts chemical energy to electrical energy
- A **resistor** causes a voltage drop — it converts electrical energy to heat

In an ideal circuit with no internal resistance:

EMF=Sum of voltage drops around the loop

This is the foundation of **Kirchhoff's Voltage Law**, which we will study in detail later.

Worked Example

Example 1.1: A battery does 48 J of work to move 6 C of charge from its negative terminal to its positive terminal. What is the EMF of the battery?

Given: $W=48 \text{ J}$, $Q=6 \text{ C}$

Required: EMF (V)

Formula: $V=W/Q$

Substitution: $V=48/6$

Answer: $V=8 \text{ V}$

Engineering Interpretation: Each coulomb of charge gains 8 joules of energy as it passes through the battery.

Worked Numerical Problem

Problem 1.1 (Level 1): How much energy is transferred when 15 mC of charge moves through a potential difference of 12 V?

Given: $Q=15 \text{ mC}=15 \times 10^{-3} \text{ C}=0.015 \text{ C}$, $V=12 \text{ V}$

Required: Energy transferred (W)

Principle: Definition of voltage

Formula: $V=W/Q \Rightarrow W=VQ$

Substitution: $W=12 \times 0.015$

Calculation: $W=0.18 \text{ J} = 180 \text{ mJ}$

Final Answer: $W=180 \text{ mJ}$

Unit Check: $V \times C = (\text{J/C}) \times C = \text{J} \checkmark$

Engineering Interpretation: 180 millijoules of energy is transferred as 15 millicoulombs of charge moves through a 12 V potential difference.

Common Mistake: Forgetting to convert mC to C before substituting. $15 \text{ mC} \neq 15 \text{ C}$.

NEC Exam Traps

1. **Confusing EMF and terminal voltage:** A real battery with internal resistance r has terminal voltage $V_T = \mathcal{E} - Ir$, which is less than EMF when current flows.
2. **Sign of voltage:** Voltage can be positive or negative depending on the reference direction. In KVL, getting signs wrong is the most common error.
3. **Voltage at a point:** Always ask "with respect to what?" There is no absolute voltage — only potential differences.

Engineering Intuition

Voltage is like the height difference in a waterfall. Water (charge) flows from high to low. The greater the height difference (voltage), the more energy the water (charge) carries. A pump (battery) lifts water back up (provides EMF), while the waterfall (resistor) converts that potential energy into kinetic energy and heat (voltage drop).

1.1.3 ELECTRIC CURRENT

Introduction

While voltage is the "cause," current is the "effect." Electric current is the flow of electric charges through a conductor. It is the quantity that actually does useful work in circuits — it heats filaments, spins motors, charges capacitors, and carries signals.

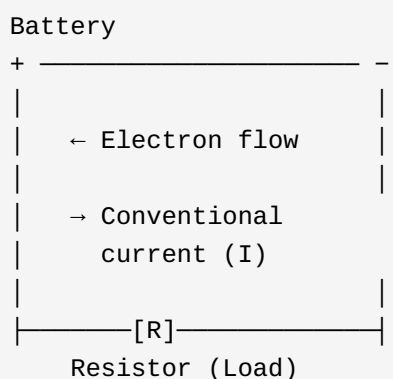
Why Is It Important?

- Current through a resistor determines heat dissipation ($P=I^2R$)
- Current determines the magnetic field around a conductor
- Circuit protection (fuses, breakers) is based on current limits
- Kirchhoff's Current Law is fundamental to circuit analysis
- Signal processing deals with time-varying currents

Basic Concept

When a voltage (potential difference) is applied across a conductor, the free electrons inside the conductor experience an electric force and begin to drift from the region of lower potential (negative terminal) toward the region of higher potential (positive terminal).

This ordered movement of charges constitutes **electric current**.



Conventional Current vs Electron Flow

CONVENTION	DIRECTION
Conventional current	From positive (+) to negative (-) terminal externally
Electron flow	From negative (-) to positive (+) terminal externally

In circuit analysis and all engineering calculations, we use **conventional current direction** unless explicitly stated otherwise. This is a universal convention — do NOT mix up the two in NEC exam problems.

Definition

Electric current is the rate of flow of electric charge through a cross-section of a conductor.

$$I = Q/t \text{ (for steady/DC current)}$$

$$i = dq/dt \text{ (general/instantaneous current)}$$

Physical Meaning

Current tells you **how many coulombs of charge pass through a point per second**.

If $I = 1 \text{ A}$, it means 6.242×10^{18} electrons pass through any cross-section of the conductor every second.

Mathematical Formulation

For constant (DC) current:

$$I = Q/t$$

For time-varying current:

$$i(t) = dq/dt$$

Conversely, charge from current:

$$Q = \int_{t_1}^{t_2} i(t) dt$$

For constant current: $Q = I \times t$

SYMBOL	MEANING	SI UNIT
I or i	Electric current	Ampere (A)
Q or q	Electric charge	Coulomb (C)
t	Time	Second (s)

Dimensional Check

$$[I] = [Q]/[t] = Cs = A$$

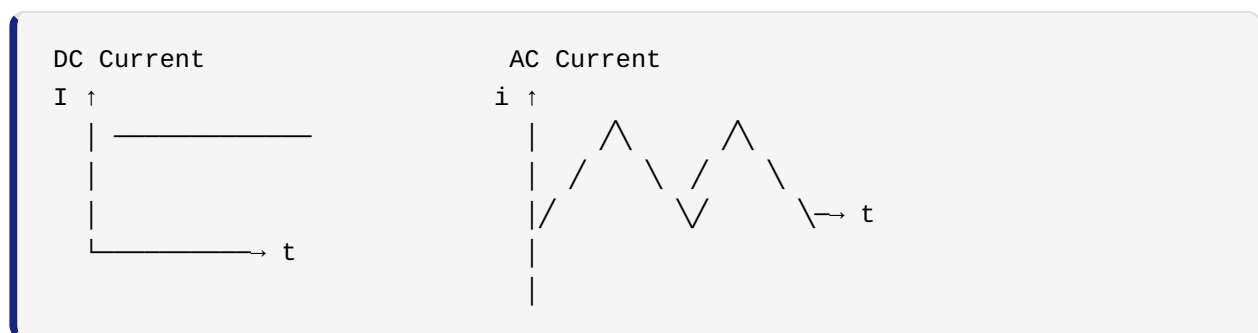
1 Ampere = 1 Coulomb per Second

Common Prefixes

PREFIX	SYMBOL	VALUE	TYPICAL USE
Milliampere	mA	10 ⁻³ A	Electronic circuits
Microampere	μA	10 ⁻⁶ A	Sensor circuits, leakage
Nanoampere	nA	10 ⁻⁹ A	Very sensitive instruments
Kiloampere	kA	10 ³ A	Power systems, lightning

Types of Current

TYPE	SYMBOL	DESCRIPTION	WAVEFORM
Direct Current (DC)	I	Constant magnitude, unidirectional	Straight horizontal line
Alternating Current (AC)	$i(t)$	Varies sinusoidally with time, reverses direction periodically	Sine wave
Pulsating DC	—	Varies in magnitude but does not reverse direction	Varying above zero



Drift Velocity (Supporting Concept)

Although individual electrons move randomly at high speeds ($\sim 10^6$ m/s), their net directed motion under an applied electric field is very slow. This net velocity is called **drift velocity** (v_d).

$$I = nAvde$$

SYMBOL	MEANING	UNIT
n	Number density of free electrons	m^{-3}
A	Cross-sectional area of conductor	m^2
vd	Drift velocity	m/s
e	Charge of electron	C

For copper, v_d is typically on the order of 10^{-4} m/s (a fraction of a millimeter per second!). Yet the **signal** (electric field) propagates at nearly the speed of light, which is why a light turns on almost instantly when you flip a switch.

Worked Example

Example 1.2: A current of 2 A flows through a wire for 5 minutes. How much charge has passed through the wire?

Given: $I = 2$ A, $t = 5$ min = 300 s

Required: Charge Q

Formula: $Q = I \times t$

Substitution: $Q = 2 \times 300 = 600$ C

Answer: $Q = 600$ C

Common Mistake: Forgetting to convert minutes to seconds. $t = 5$ min $\neq$ 5 s.

Worked Numerical Problem

Problem 1.2 (Level 2): The current through a conductor varies as $i(t) = 3t^2 + 2t$ amperes. Find the total charge that flows between $t = 1$ s and $t = 3$ s.

Given: $i(t) = 3t^2 + 2t$ A, $t_1 = 1$ s, $t_2 = 3$ s

Required: Total charge Q

Principle: Charge is the integral of current over time

Formula: $Q = \int_{t_1}^{t_2} i(t) dt$

Substitution:

$$Q = \int_{13}^{33} (3t^2 + 2t) dt$$

Calculation:

$$Q = [t^3 + t^2]_{13}^{33}$$

$$Q = (33^3 + 33^2) - (13^3 + 13^2)$$

$$Q = (27+9) - (1+1)$$

$$Q = 36 - 2 = 34 \text{ C}$$

Final Answer: $Q = 34 \text{ C}$

Unit Check: $\int A \cdot s = C \checkmark$

Engineering Interpretation: 34 coulombs of charge flowed through the conductor in the 2-second interval. The current is increasing with time (due to the t^2 and t terms), so most of this charge flows during the latter part of the interval.

NEC Exam Traps

1. **Unit conversion:** Always convert time to seconds before calculation.
2. **Current direction:** In circuit analysis, if you assume a current direction and get a negative answer, it means the actual current flows opposite to your assumed direction — it does NOT mean your analysis is wrong.
3. **DC vs AC current:** In DC, $I = Q/t$ directly. For AC or time-varying current, you must integrate.



Engineering Intuition

Current is like the flow rate of water in a pipe. A 1-ampere current means 1 coulomb of charge passes any point per second. A thicker pipe (larger conductor cross-section) doesn't mean more current unless you increase the pressure (voltage). Current depends on voltage AND resistance.

1.1.4 RESISTANCE AND OHM'S LAW

Introduction

Resistance is the property of a material that **opposes** the flow of electric current. It is the "friction" of the electrical world. Understanding resistance is essential because every real conductor has some resistance, and this resistance determines how much current flows for a given voltage.

Ohm's Law — the relationship between voltage, current, and resistance — is the single most important equation in electrical engineering.

Why Is It Important?

- Ohm's Law is the foundation of ALL circuit analysis
 - Resistance determines current for a given voltage
 - Resistance determines power dissipation (heat)
 - Understanding resistance is needed for: series/parallel circuits, voltage dividers, current dividers, network theorems, impedance (AC), and transistor biasing
 - Every NEC exam will have problems requiring Ohm's Law
-

RESISTANCE

Basic Concept

When current flows through a conductor, the moving electrons collide with the fixed atoms of the conductor material. These collisions:

- Impede the flow of electrons (resistance)
- Convert electrical energy to heat (Joule heating)

Different materials offer different levels of opposition:

- **Copper**: Low resistance → Good conductor

- **Glass**: Extremely high resistance → Insulator
- **Silicon**: Moderate resistance → Semiconductor

Definition

Resistance is the property of a material that opposes the flow of electric current through it. It is measured in **ohms (Ω)**.

Mathematically, resistance is the ratio of voltage across an element to the current through it.

$$R=VI$$

1 Ohm: A conductor has a resistance of $1\ \Omega$ if a potential difference of $1\ \text{V}$ across it produces a current of $1\ \text{A}$.

Physical Meaning

Resistance represents **how difficult it is for current to flow through a material**. High resistance means:

- For the same voltage, less current flows
- More energy is converted to heat per unit charge
- The material is a poorer conductor

Factors Affecting Resistance

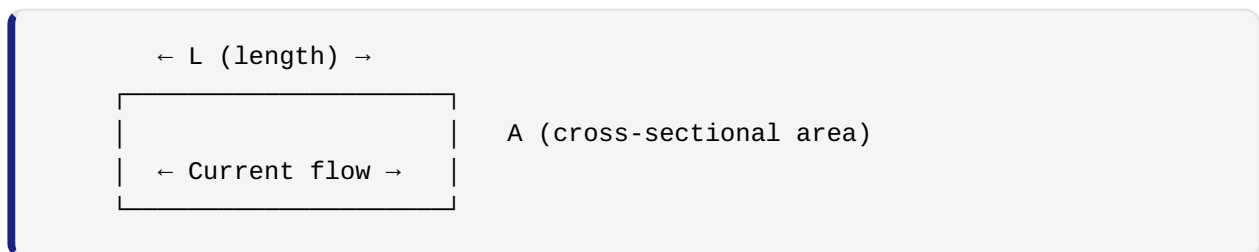
The resistance of a conductor depends on:

$$R=\rho LA$$

SYMBOL	MEANING	SI UNIT
R	Resistance	Ohm (Ω)
ρ	Resistivity (material property)	$\Omega \cdot \text{m}$
L	Length of conductor	m
A	Cross-sectional area	m^2

Derivation of $R = \rho L / A$

Consider a uniform cylindrical conductor:



Physical reasoning:

1. **Length (L):** A longer conductor means electrons must travel farther and undergo more collisions. Therefore, $R \propto L$.
2. **Area (A):** A larger cross-section provides more "lanes" for electrons. Therefore, $R \propto 1/A$.
3. **Resistivity (ρ):** Different materials have different atomic structures and electron densities. This material-dependent constant captures the intrinsic opposition of the material.

Combining: $R = \rho L / A$

Important Resistivity Values

MATERIAL	RESISTIVITY P ($\Omega \cdot m$) AT 20°C	CLASSIFICATION
Silver	1.59×10^{-8}	Conductor
Copper	1.68×10^{-8}	Conductor
Aluminum	2.65×10^{-8}	Conductor
Tungsten	5.6×10^{-8}	Conductor
Silicon	6.4×10^2	Semiconductor
Glass	1010 to 1014	Insulator
Rubber	$\sim 10^{13}$	Insulator

Conductance

The reciprocal of resistance is **conductance**:

$$G = 1/R = A/\rho L = \sigma A/L$$

SYMBOL	MEANING	SI UNIT
G	Conductance	Siemens (S) or mho (Ω)
σ	Conductivity ($=1/\rho$)	S/m

Temperature Effect on Resistance

For most metals, resistance **increases** with temperature:

$$R_T = R_0[1 + \alpha(T - T_0)]$$

SYMBOL	MEANING	UNIT
R_T	Resistance at temperature T	Ω
R_0	Resistance at reference temperature T_0	Ω
α	Temperature coefficient of resistance	$^{\circ}\text{C}^{-1}$
T	Operating temperature	$^{\circ}\text{C}$
T_0	Reference temperature (usually 20°C)	$^{\circ}\text{C}$

- **Metals:** $\alpha > 0$ (positive temperature coefficient — PTC)
- **Semiconductors:** $\alpha < 0$ (negative temperature coefficient — NTC)
- **Alloys** (e.g., Manganin, Constantan): $\alpha \approx 0$ (used for precision resistors)

OHM'S LAW

Basic Concept

Georg Simon Ohm discovered experimentally (1827) that for many materials, the current through a conductor is **directly proportional** to the voltage across it, provided the temperature and other physical conditions remain constant.

Statement

Ohm's Law: The current flowing through a conductor is directly proportional to the potential difference across it and inversely proportional to its resistance, provided the temperature remains constant.

Mathematical Formulation

$$V = IR$$

This can be rearranged into three forms:

FORM	FORMULA	USE WHEN YOU KNOW	AND NEED
1	$V=IR$	Current and Resistance	Voltage
2	$I=VR$	Voltage and Resistance	Current
3	$R=VI$	Voltage and Current	Resistance

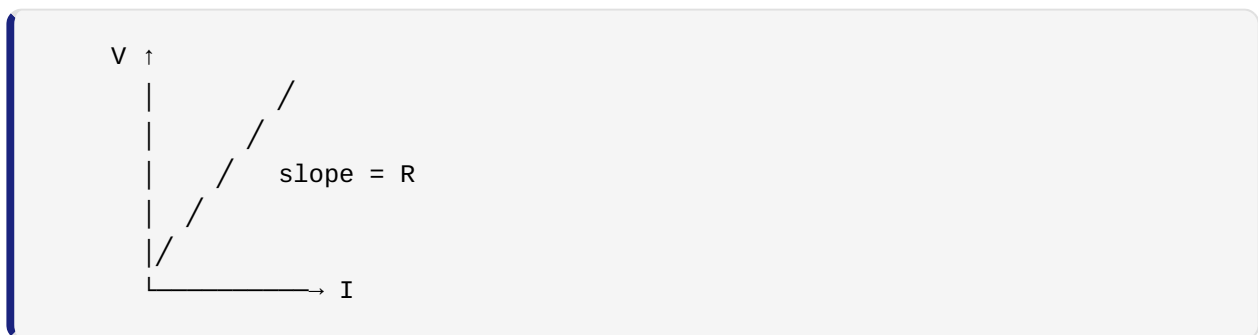
SYMBOL	MEANING	SI UNIT
V	Voltage across the element	Volt (V)
I	Current through the element	Ampere (A)
R	Resistance of the element	Ohm (Ω)

Conditions of Validity

Ohm's Law is valid when:

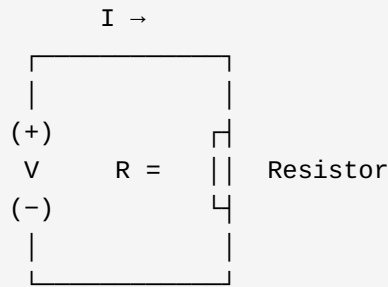
1. **Temperature is constant** (resistance changes with temperature)
2. **The material is ohmic** (linear V-I relationship)
3. **Physical conditions are constant** (no mechanical stress, constant dimensions)

V-I Characteristic of an Ohmic Resistor



The V-I characteristic of an ohmic (linear) resistor is a **straight line through the origin**. The slope equals the resistance R .

Ohm's Law — Circuit Diagram



$$V = IR$$

Note: Current enters the positive terminal of the resistor (passive sign convention)

Passive Sign Convention

In **passive sign convention**, the current enters the positive terminal of a passive element (resistor). This gives $V = +IR$. If current enters the negative terminal, then $V = -IR$. Getting this convention right is critical for KVL problems.

Physical Interpretation

Ohm's Law tells us:

1. **For fixed R :** Doubling the voltage doubles the current (linear relationship)
2. **For fixed V :** Doubling the resistance halves the current
3. **For fixed I :** Doubling the resistance doubles the required voltage

Derivation (Microscopic)

While Ohm's Law is empirical (discovered through experiment), it can be derived from the microscopic behavior of electrons in a conductor.

Starting from the drift velocity relation:

$$J = \sigma E$$

where J is current density (A/m^2), σ is conductivity (S/m), and E is electric field (V/m).

For a uniform conductor of length L and cross-section A :

$$J=IA, E=VL$$

Substituting:

$$IA=\sigma VL$$

$$I=\sigma AL \cdot V$$

$$I=VR \text{ where } R=L/\sigma A=\rho LA$$

This gives us Ohm's Law: $V=IR$, derived from the fundamental relationship between current density and electric field.

Worked Example

Example 1.3: A resistor of 470Ω has a current of 20 mA flowing through it. What is the voltage across the resistor?

Given: $R=470 \Omega$, $I=20 \text{ mA}=0.02 \text{ A}$

Required: Voltage V

Formula: $V=IR$

Substitution: $V=0.02 \times 470$

Answer: $V=9.4 \text{ V}$

Worked Numerical Problem

Problem 1.3 (Level 2): A copper wire is 200 m long with a cross-sectional area of 2 mm^2 . The resistivity of copper is $1.68 \times 10^{-8} \Omega \cdot \text{m}$. (a) Find the resistance of the wire. (b) If a voltage of 5 V is applied across the wire, find the current. (c) Find the power dissipated.

Given: $L=200 \text{ m}$, $A=2 \text{ mm}^2=2 \times 10^{-6} \text{ m}^2$, $\rho=1.68 \times 10^{-8} \Omega \cdot \text{m}$, $V=5 \text{ V}$

(a) Find Resistance:**Formula:** $R = \rho LA$ **Substitution:** $R = 1.68 \times 10^{-8} \times 2002 \times 10^{-6}$ **Calculation:**

$$R = 3.36 \times 10^{-8} \times 10^{-6} = 1.68 \, \Omega$$

(b) Find Current:**Formula:** $I = VR$ **Substitution:** $I = 51.68$ **Calculation:** $I = 2.976 \, \text{A} \approx 2.98 \, \text{A}$ **(c) Find Power:** (Using concept from next section)

$$P = VI = 5 \times 2.976 = 14.88 \, \text{W}$$

Unit Check: $[\Omega \cdot \text{m}][\text{m}][\text{m}^2] = \Omega \, \checkmark$ **Common Mistake:** Forgetting to convert mm^2 to m^2 . $2 \, \text{mm}^2 = 2 \times 10^{-6} \, \text{m}^2$, NOT $2 \times 10^{-3} \, \text{m}^2$.**Area conversion is the #1 source of errors in resistance calculations:**- $1 \, \text{mm}^2 = 10^{-6} \, \text{m}^2$ (NOT 10^{-3})- $1 \, \text{cm}^2 = 10^{-4} \, \text{m}^2$ (NOT 10^{-2})Remember: $1 \, \text{mm} = 10^{-3} \, \text{m}$, so $1 \, \text{mm}^2 = (10^{-3})^2 = 10^{-6} \, \text{m}^2$ **Numerical Problem — Level 3 (NEC Challenge)**

Problem 1.4: A heating element made of nichrome wire has a resistance of $50 \, \Omega$ at 20°C . The temperature coefficient of nichrome is $\alpha = 0.0004 \, ^\circ\text{C}^{-1}$. If the element operates at 520°C , find:

(a) The resistance at operating temperature

- (b) The current drawn from a 220 V supply at operating temperature
 (c) The percentage change in current compared to the current at 20°C

Given: $R_0 = 50 \, \Omega$, $T_0 = 20^\circ\text{C}$, $\alpha = 0.0004 \, ^\circ\text{C}^{-1}$, $T = 520^\circ\text{C}$, $V = 220 \, \text{V}$

(a) Resistance at 520°C:

$$R_T = R_0[1 + \alpha(T - T_0)]$$

$$R_{520} = 50[1 + 0.0004(520 - 20)]$$

$$R_{520} = 50[1 + 0.0004 \times 500]$$

$$R_{520} = 50[1 + 0.2] = 50 \times 1.2$$

$$R_{520} = 60 \, \Omega$$

(b) Current at operating temperature:

$$I_{520} = \frac{V}{R_{520}} = \frac{220}{60} = 3.667 \, \text{A}$$

(c) Percentage change in current:

Current at 20°C: $I_{20} = \frac{220}{50} = 4.4 \, \text{A}$

$$\text{Percentage change} = \frac{I_{20} - I_{520}}{I_{20}} \times 100\%$$

$$= \frac{4.4 - 3.667}{4.4} \times 100\% = 0.1667 \times 100\% \approx 16.67\%$$

Engineering Interpretation: The resistance increased by 20% (from 50 Ω to 60 Ω) due to temperature rise, causing the current to decrease by about 16.7%. This is why incandescent bulbs draw a large surge current when first turned on (filament is cold = low resistance) and then the current settles down as the filament heats up.

Applications

1. **Resistors in electronics:** Current limiting, voltage division, biasing
2. **Heating elements:** Ohmic heating (electric stoves, heaters)
3. **Sensors:** Thermistors (NTC, PTC), strain gauges
4. **Fuses:** Designed to melt (break circuit) when current exceeds rated value

Ohmic vs Non-Ohmic Devices

PROPERTY	OHMIC DEVICE	NON-OHMIC DEVICE
V-I curve	Straight line through origin	Non-linear / Curved
Resistance	Constant	Varies with V or I
Obeys Ohm's Law?	Yes	Not strictly
Examples	Metal resistors, copper wire	Diodes, transistors, filament lamps

Ohmic (Resistor)



Non-Ohmic (Diode)



Non-ohmic devices do NOT violate Ohm's Law. Ohm's Law ($V=IR$) still applies instantaneously — the key difference is that R is not constant. At any given operating point, we can define a **dynamic resistance** $r=dV/dI$.

⚠️ NEC Exam Traps — Ohm's Law and Resistance

1. **Using $R=V/I$ for a diode:** This gives the static (DC) resistance, NOT the dynamic (AC) resistance. The dynamic resistance is $r_d=dV/dI$, which is different.

2. **Applying Ohm's Law across a voltage source:** A voltage source has (ideally) zero resistance. $R=V/I$ applied to the source gives you the load resistance, NOT the source resistance.
3. **Unit traps in resistance calculation:**
 - $\text{mm}^2 \rightarrow \text{m}^2$: multiply by 10^{-6}
 - $\text{cm} \rightarrow \text{m}$: multiply by 10^{-2}
 - $\text{k}\Omega \rightarrow \Omega$: multiply by 10^3
4. **Temperature effects:** NEC may give resistance at one temperature and ask for current at another. You must use the temperature formula.
5. **Sign convention:** Current ENTERS the positive terminal of a resistor. If you define the current the other way, you must use $V=-IR$.



Engineering Intuition — Ohm's Law

Ohm's Law is not just a formula — it is a way of thinking.

When you see a resistor in a circuit, immediately think:

- "What voltage is across it?"
- "What current flows through it?"
- "How much power does it dissipate?"

These three questions, answerable by $V=IR$, $I=V/R$, and $P=I^2R=V^2/R=VI$, are the core of circuit analysis. Every network theorem (Thevenin, Norton, Superposition) ultimately reduces a complex circuit to a simple Ohm's Law problem.

Ohm's Law — Quick Revision Triangle

A useful memory aid:



Cover what you want:

- Cover V: $V = I \times R$
- Cover I: $I = V / R$
- Cover R: $R = V / I$

1.1.5 ELECTRIC POWER

Introduction

Power is the rate at which energy is transferred or converted. In electrical circuits, power tells us how fast a circuit element is converting electrical energy into another form (heat, light, mechanical energy, etc.) or how fast a source is supplying energy to the circuit.

Why Is It Important?

- Power determines the **rating** of every electrical device
- Power dissipation determines **heat generation** (thermal management)
- Power efficiency determines **energy costs** and **battery life**
- Power calculations are needed for: component selection, safety analysis, efficiency calculations, and billing

Definition

Electric power is the rate at which electrical energy is transferred by a circuit element.

$$P = \frac{W}{t} = \frac{\text{Energy}}{\text{Time}}$$

Mathematical Formulation

Fundamental Formula

$$P = VI$$

Derivation

Starting from the definitions of voltage and current:

$$V=WQ \Rightarrow W=VQ$$

$$I=Qt \Rightarrow Q=It$$

Substituting:

$$W=V \times It = VIt$$

$$P=Wt=VItt=VI$$

Derived Forms (using Ohm's Law)

Since $V=IR$:

$$P=VI=(IR) \times I = I^2R$$

Since $I=V/R$:

$$P=VI=V \times V/R = V^2/R$$

Complete Power Formula Set

FORMULA	VARIABLES KNOWN	BEST USED WHEN
$P=VI$	Voltage and Current	General case
$P=I^2R$	Current and Resistance	Current is known/given
$P=V^2/R$	Voltage and Resistance	Voltage is known/given

SYMBOL	MEANING	SI UNIT
P	Electric power	Watt (W)
V	Voltage	Volt (V)
I	Current	Ampere (A)
R	Resistance	Ohm (Ω)

Dimensional Check

$$[P] = [V][I] = JC \times Cs = Js = W$$

Common Prefixes

PREFIX	VALUE	TYPICAL USE
mW (milliwatt)	10 ⁻³ W	Electronic ICs, sensors
W (watt)	100 W	Light bulbs, small appliances
kW (kilowatt)	10 ³ W	Heaters, motors, homes
MW (megawatt)	10 ⁶ W	Power stations, industrial
GW (gigawatt)	10 ⁹ W	National power grids

Physical Meaning

Power consumed by a resistor: Electrical energy is irreversibly converted to heat. This is called **Joule heating** or **ohmic loss**.

Power delivered by a source: The source (battery, generator) converts another form of energy (chemical, mechanical) into electrical energy and supplies it to the circuit.

Sign Convention for Power

CONDITION	POWER	MEANING
Current enters (+) terminal	$P > 0$	Element absorbs (consumes) power
Current enters (-) terminal	$P < 0$	Element delivers (supplies) power

Absorbing power:

$I \rightarrow$
 $(+) \text{---} (-)$
 $P = +VI$
 (Resistor)

Delivering power:

$I \rightarrow$
 $(-) \text{---} (+)$
 $P = -VI$
 (Battery)

Worked Numerical Problem

Problem 1.5 (Level 2): An electric heater is rated at 1500 W, 220 V. Find:

- The current drawn
- The resistance of the heating element
- The cost of running the heater for 8 hours if electricity costs NPR 10 per kWh

Given: $P=1500$ W, $V=220$ V, $t=8$ hours, cost = NPR 10/kWh

(a) Current:

$$I = \frac{P}{V} = \frac{1500}{220} = 6.82 \text{ A}$$

(b) Resistance:

$$R = \frac{V^2}{P} = \frac{220^2}{1500} = 32.27 \text{ } \Omega$$

Verification: $P = I^2 R = 6.82^2 \times 32.27 = 46.51 \times 32.27 \approx 1500 \text{ W}$ ✓

(c) Cost:

$$\text{Energy consumed} = P \times t = 1500 \times 8 = 12,000 \text{ Wh} = 12 \text{ kWh}$$

$$\text{Cost} = 12 \times 10 = \text{NPR } 120$$

Engineering Interpretation: The heater draws about 6.82 A from a 220 V supply. This is within the typical 16 A socket rating in Nepal. The heating element resistance is about 32 Ω . Running for 8 hours consumes 12 kWh of energy.

1.1.6 ELECTRIC ENERGY

Introduction

While power tells us the *rate* of energy transfer, energy tells us the *total amount* of work done or energy consumed over a period of time. Energy is what you pay for in your electricity bill.

Definition

Electric energy is the total amount of work done by or on electric charges over a period of time.

$$W = Pt$$

Mathematical Formulation

From power:

$$W = Pt = VIt = I^2Rt = V^2Rt$$

SYMBOL	MEANING	SI UNIT
W (or E)	Electric energy	Joule (J)
P	Power	Watt (W)
t	Time	Second (s)

Commercial Unit

In power systems, the joule is too small for practical use. The commercial unit is:

$$1 \text{ kWh} = 1 \text{ kilowatt} \times 1 \text{ hour} = 1000 \times 3600 = 3.6 \times 10^6 \text{ J} = 3.6 \text{ MJ}$$

kWh is a unit of ENERGY, not power. This is one of the most common confusions.

- **kW** → Power (rate)
- **kWh** → Energy (total amount)

Comparison: Power vs Energy

PROPERTY	POWER	ENERGY
Definition	Rate of energy transfer	Total energy transferred
Formula	$P=VI$	$W=Pt=VIt$
SI Unit	Watt (W)	Joule (J)
Commercial Unit	kW, MW	kWh
Analogy	Speed of water flow	Total water flowed
Time dependency	Instantaneous	Cumulative over time
What it measures	How fast energy is used	How much energy is used
What you pay for	No (indirectly through demand charges)	Yes (energy charges)



NEC Exam Traps — Power and Energy

1. **Confusing power and energy:** The question may ask for energy but give power, or vice versa. Read carefully.
2. **Unit mismatch:** If power is in kW and time is in hours, energy is directly in kWh. But if time is in seconds, convert: $E(\text{kWh}) = P(\text{W}) \times t(\text{s}) / 3.6 \times 10^6$.

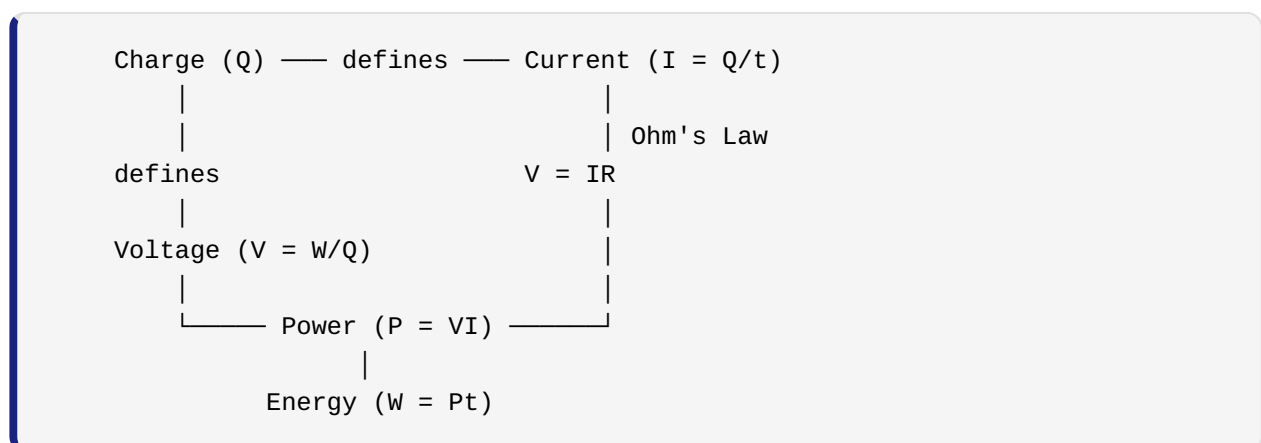
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3. **"Power consumed by a 100 W bulb for 5 hours"**: The power is always 100 W. The ENERGY consumed is 500 Wh = 0.5 kWh.
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SECTION 1.1 PART 1 — SUMMARY AND KEY EQUATIONS

Key Concepts Covered

#	CONCEPT	KEY EQUATION
1	Electric Charge	$Q = ne$
2	Voltage (Potential Difference)	$V = W/Q$
3	Electric Current	$I = Q/t$ or $i = dq/dt$
4	Resistance	$R = \rho L/A$
5	Ohm's Law	$V = IR$
6	Temperature Effect	$R_T = R_0[1 + \alpha(T - T_0)]$
7	Electric Power	$P = VI = I^2R = V^2/R$
8	Electric Energy	$W = Pt = VIt$

Important Relationships Map



Comparison Table: Voltage vs Current

PROPERTY	VOLTAGE	CURRENT
Definition	Energy per unit charge	Charge per unit time
Symbol	V	I
SI Unit	Volt (V)	Ampere (A)
Measured	Between two points (across)	At a point (through)
Instrument	Voltmeter (connected in parallel)	Ammeter (connected in series)
Analogy	Water pressure (height)	Water flow rate
Cause/Effect	Cause	Effect
In series circuit	Divides across elements	Same through all elements
In parallel circuit	Same across all elements	Divides among branches

Comparison Table: Power vs Energy

PROPERTY	POWER	ENERGY
Definition	Rate of energy transfer	Total energy transferred
Formula	$P=VI$	$W=Pt$
SI Unit	Watt (W) = J/s	Joule (J)
Commercial Unit	kW	kWh
Relationship	$P=dW/dt$	$W=\int Pdt$

What's Next

Section 1.1 — Part 2 will cover:

- Conducting and Insulating Materials
- Series Electric Circuits (with voltage divider)
- Parallel Electric Circuits (with current divider)
- Star (Y) and Delta (Δ) Connections
- Star-Delta and Delta-Star Conversion (with derivation)

These build directly on the V, I, R, P, and E concepts established in this part.

Chapter 1: Basic Concept (Section 1.1 — Part 2)

1.1.8 Conducting and Insulating Materials

Introduction

Every material interacts with electric charge differently. While some materials permit easy flow of electrons (conductors), others strongly oppose it (insulators). Understanding these distinct properties is the foundation of circuit design, safety, and component selection.

Basic Concept

The conductivity of a material depends primarily on its atomic structure, specifically the behavior of its valence electrons (electrons in the outermost shell).

1. Conductors

- **Definition:** Materials that offer very little resistance to the flow of electric current.
- **Physical Meaning:** In conductors, atoms have 1 to 3 loosely bound valence electrons. At room temperature, thermal energy is enough to free these electrons from their parent atoms. These **free electrons** form an "electron gas" and drift freely when an electric field (voltage) is applied.
- **Examples:** Silver (best), Copper, Gold, Aluminum.
- **Why Metals Conduct:** The metallic bond allows electrons to move freely throughout the crystal lattice.

2. Insulators

- **Definition:** Materials that strongly resist the flow of electric current.
- **Physical Meaning:** Insulators have 5 to 8 tightly bound valence electrons. There are practically zero free electrons available for conduction. An enormous voltage (breakdown voltage) is required to rip these electrons from their atoms to force conduction.
- **Examples:** Glass, Rubber, Mica, Ceramic, Dry Air.
- **Why they don't conduct:** All valence electrons are locked in covalent or ionic bonds, leaving no charge carriers.

3. Semiconductors (Preview)

- **Definition:** Materials with electrical conductivity between that of a conductor and an insulator.
- **Physical Meaning:** At absolute zero, they act as perfect insulators. As temperature rises, some covalent bonds break, creating free electrons and holes, thus allowing some conduction. They have exactly 4 valence electrons.
- **Examples:** Silicon, Germanium.

Comparison Table

FEATURE	CONDUCTOR	INSULATOR	SEMICONDUCTOR
Valence Electrons	1, 2, or 3	5, 6, 7, or 8	Exactly 4
Resistivity (ρ)	Low ($10^{-8} \Omega \cdot m$)	Very High ($10^{12} \Omega \cdot m$)	Moderate ($10^{-3} \Omega \cdot m$)
Temperature Coefficient	Positive (Resistance increases with Temp)	Negative (Resistance drops slightly)	Negative (Resistance drops heavily)
Energy Gap	Overlapping bands	Large ($>5 \text{ eV}$)	Small ($\sim 1 \text{ eV}$)

 **NEC Exam Trap:** You might be asked what happens to the resistance of a semiconductor or insulator when heated. Remember: Conductors get *more* resistive when hot; Semiconductors and Insulators get *less* resistive when hot (Negative Temperature Coefficient).

1.1.9 Series Electric Circuits

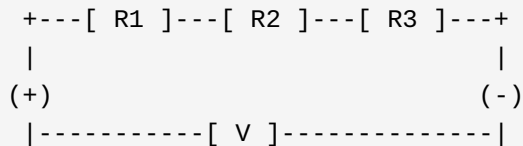
Introduction

The series circuit is the simplest way to connect components. It forms a single, undivided path for current.

Definition

A series circuit is one in which components are connected end-to-end, so that the **same current** flows through all components.

Circuit Diagram



Mathematical Formulation & Derivations

1. Current

Because there is only one path, the current is identical everywhere.

$$I_{\text{total}} = I_1 = I_2 = I_3$$

2. Total Resistance

From Ohm's Law and conservation of energy (KVL), the total voltage V provided by the source equals the sum of voltage drops across resistors.

$$V_{\text{total}} = V_1 + V_2 + V_3$$

Substitute $V = I \times R$:

$$I \cdot R_{\text{eq}} = I \cdot R_1 + I \cdot R_2 + I \cdot R_3$$

Dividing by I :

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots + R_n$$

3. Voltage Divider Rule (VDR)

In a series circuit, the total voltage divides among the resistors in direct proportion to their resistance.

$$V_k = I \cdot R_k = (V_{\text{total}} / R_{\text{eq}}) R_k$$

$$V_k = V_{\text{total}} \times R_k / R_{\text{eq}}$$

SYMBOL	MEANING	SI UNIT
V_k	Voltage drop across resistor k	Volts (V)
R_k	Resistance of resistor k	Ohms (Ω)
R_{eq}	Total equivalent series resistance	Ohms (Ω)

4. Power

Total power dissipated is the sum of individual powers.

$$P_{\text{total}} = P_1 + P_2 + P_3 = I^2 R_1 + I^2 R_2 + I^2 R_3$$



Engineering Intuition: If you string a bunch of old Christmas lights in series and one burns out (breaks the circuit), they all go dark because the single current path is broken.

Worked Numerical (Level 2)

Given: Three resistors, $R_1 = 10 \Omega$, $R_2 = 20 \Omega$, and $R_3 = 30 \Omega$, are connected in series across a 120V supply.

Required:

1. Total equivalent resistance
2. Circuit current
3. Voltage drop across R_2

Principle: Series circuit properties and Voltage Divider Rule.

Calculation:

1. $R_{\text{eq}} = 10 + 20 + 30 = 60 \Omega$
2. $I = V / R_{\text{eq}} = 120 / 60 = 2 \text{ A}$
3. Using VDR for R_2 :

$$V_2 = 120 \times \frac{20}{60} = 120 \times \frac{1}{3} = 40 \text{ V}$$

.

(Check: $V_2 = I \times R_2 = 2 \times 20 = 40 \text{ V}$).

Interpretation: The largest resistor (30Ω) will have the largest voltage drop (60 V), keeping true to proportional division.

1.1.10 Parallel Electric Circuits

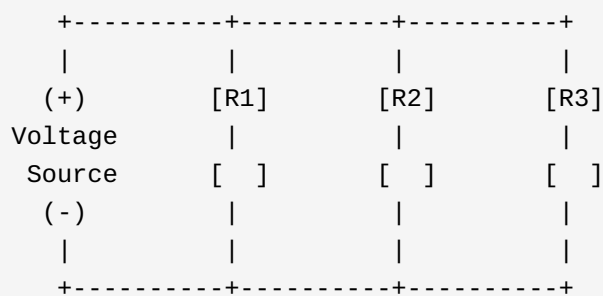
Introduction

In a parallel circuit, components are connected across the same common nodes, providing multiple paths for the current to flow.

Definition

A parallel circuit is one in which all components share the same two nodes, meaning the **same voltage** appears across all components.

Circuit Diagram



Mathematical Formulation & Derivations

1. Voltage

The voltage across every parallel branch is identical.

$$V_{\text{total}} = V_1 = V_2 = V_3$$

2. Total Resistance

By KCL (conservation of charge), total current is the sum of branch currents:

$$I_{\text{total}} = I_1 + I_2 + I_3$$

Substitute $I = V/R$:

$$V_{\text{Req}} = V_{R1} + V_{R2} + V_{R3}$$

Dividing by V :

$$1/R_{\text{eq}} = 1/R_1 + 1/R_2 + 1/R_3 + \dots + 1/R_n$$

Special Case (Two Resistors):

For just two resistors in parallel:

$$R_{\text{eq}} = \frac{R_1 \times R_2}{R_1 + R_2}$$

(Product over Sum)

3. Current Divider Rule (CDR)

Current divides inversely proportional to resistance (path of least resistance gets most current).

$$I_k = V/R_k = I_{\text{total}} \times \frac{R_{\text{eq}}}{R_k} = I_{\text{total}} \times \frac{R_{\text{eq}}}{R_k}$$

For exactly two resistors in parallel:

$$I_1 = I_{\text{total}} \times \frac{R_2}{R_1 + R_2}$$

$$I_2 = I_{\text{total}} \times \frac{R_1}{R_1 + R_2}$$

(Notice: To find I_1 , use R_2 in the numerator!)

4. Power

Total power is still the sum of individual powers.

$$P_{\text{total}} = P_1 + P_2 + P_3 = \frac{V^2}{R_1} + \frac{V^2}{R_2} + \frac{V^2}{R_3}$$

Common Mistake: Students often use $I_1 = I_{\text{total}} R_1 / (R_1 + R_2)$ which is WRONG. Remember current favors the *other* path if your resistance is high, so use the *opposite* resistance in the numerator.

Series vs Parallel Comparison Table

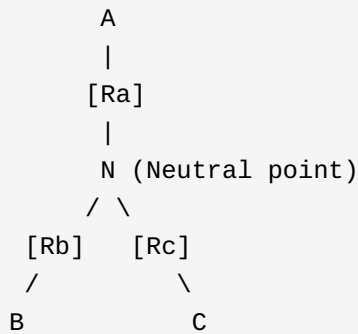
PROPERTY	SERIES	PARALLEL
Current	Same through all elements	Divides among branches
Voltage	Divides among elements	Same across all branches
Total Resistance	Increases ($R_{\text{eq}} > R_{\text{largest}}$)	Decreases ($R_{\text{eq}} < R_{\text{smallest}}$)
Open Circuit Effect	Entire circuit stops functioning	Other branches continue functioning
Household Wiring	Seldom used (except switches)	Standard for all appliances

1.1.11 Star (Y) Connection

What is a Star/Y Connection?

A Star (or Wye, Y) connection is formed when one end of three components (typically resistors or coils) are joined together at a common point called the **neutral point** or **star point**, leaving the other three ends free to connect to the external circuit.

Circuit Diagram



Characteristics

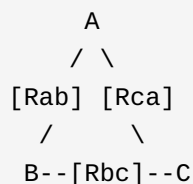
- Symmetrical appearance resembles the letter Y.
- Accessible neutral point (N) allows for 3-phase, 4-wire systems (discussed later in AC).
- Between any two terminals (e.g., A and B), two resistors are in series if the third terminal is open.

1.1.12 Delta (Δ) Connection

What is a Delta Connection?

A Delta (or Mesh, Δ) connection is formed when three components are connected end-to-end to form a closed loop. The three connection points (nodes) are used to connect to the external circuit.

Circuit Diagram



Characteristics

- Resembles the Greek letter Delta (Δ).
- No neutral point exists.
- Between any two terminals (e.g., A and B), there is one resistor (R_{ab}) in parallel with the series combination of the other two ($R_{bc}+R_{ca}$).

1.1.13 Star-Delta and Delta-Star Conversion

Why we need to convert

In complex bridge networks (like a Wheatstone bridge that isn't balanced), resistors are neither purely in series nor in parallel. Star-Delta transformations allow us to redraw these stubborn geometries into simple series-parallel combinations to calculate equivalent resistance.

1. Delta to Star Conversion ($\Delta \rightarrow Y$)

To replace a Delta network (R_{ab}, R_{bc}, R_{ca}) with an equivalent Star network (R_a, R_b, R_c), the resistance between any two terminals must be identical in both configurations.

Derivation Principle:

Resistance between A and B in Star: $R_a + R_b$

Resistance between A and B in Delta: $R_{ab} \parallel (R_{bc} + R_{ca})$

Equating these for all three terminal pairs and solving algebraically yields:

Formulas:

$$R_a = \frac{R_{ab}R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_b = \frac{R_{ab}R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_c = \frac{R_{bc}R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

Rule to remember: The Star resistor connected to a terminal is the product of the two Delta resistors connected to that same terminal, divided by the sum of all three Delta resistors.

2. Star to Delta Conversion ($Y \rightarrow \Delta$)

Formulas:

$$R_{ab} = R_a + R_b + \frac{R_a R_b}{R_c} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c}$$

$$R_{bc} = R_b + R_c + \frac{R_b R_c}{R_a} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_a}$$

$$R_{ca} = R_c + R_a + \frac{R_c R_a}{R_b} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_b}$$

Rule to remember: The Delta resistor between two terminals is the sum of all pair-wise products of Star resistors, divided by the Star resistor connected to the *opposite* (third) terminal.

Special Case: Equal Resistances

If all resistors are equal ($R_a = R_b = R_c = R_Y$) and ($R_{ab} = R_{bc} = R_{ca} = R_{\Delta}$):

$$R_{\Delta} = 3 \times R_Y$$

$$R_Y = \frac{R_{\Delta}}{3}$$

⚠ NEC Exam Trap: A frequent multiple-choice question: "If three 10 Ω resistors are connected in Delta, what is the equivalent Star resistance?" Answer: 10/3 = 3.33 Ω . Students often mistakenly multiply by 3 instead of divide. Remember: Delta values are always LARGER than their equivalent Star values (by a factor of 3 for identical resistors).

Worked Numerical (Level 3 - NEC Challenge)

Given: A bridge circuit where a Delta connection exists between terminals A, B, and C with $R_{ab}=10\ \Omega$, $R_{bc}=20\ \Omega$, $R_{ca}=30\ \Omega$.

Required: Find the equivalent Star connection resistors R_a , R_b , R_c .

Formula: $R_{star} = \text{Product of adjacent } R \Delta \Sigma R \Delta$

Calculation:

$$1. \Sigma R \Delta = 10 + 20 + 30 = 60\ \Omega$$

2. R_a (connected to node A, between $10\ \Omega$ and $30\ \Omega$):

$$R_a = \frac{10 \times 30}{60} = 5\ \Omega$$

1. R_b (connected to node B, between $10\ \Omega$ and $20\ \Omega$):

$$R_b = \frac{10 \times 20}{60} = 3.33\ \Omega$$

2. R_c (connected to node C, between $20\ \Omega$ and $30\ \Omega$):

$$R_c = \frac{20 \times 30}{60} = 10\ \Omega$$

Final Answer: $R_a=5\ \Omega$, $R_b=3.33\ \Omega$, $R_c=10\ \Omega$.

1.1.14 Kirchhoff's Current Law (KCL)

Statement

Kirchhoff's First Law states that the algebraic sum of all currents entering and leaving a node (junction) must equal zero. Alternatively, the sum of currents entering a node equals the sum of currents leaving the node.

Physical Basis

Conservation of Charge: Charge cannot be created or destroyed at a junction. Whatever electric charge flows in must flow out.

Sign Convention

- Currents ENTERING the node: POSITIVE (+)

Sign Convention for Loop Traversal

When walking around a loop in a chosen direction (e.g., clockwise):

1. **Battery:** Going from (-) to (+) is a voltage RISE (+V). Going from (+) to (-) is a voltage DROP (-V).
2. **Resistor:** Going *with* the current direction is a voltage DROP (-IR). Going *against* the current is a voltage RISE (+IR).

Mathematical Formulation

$$\sum_{n=1}^N V_n = 0$$

around any closed loop.

Worked Numerical (Mesh Analysis concept using KVL)

Given: A single loop circuit with a 12V battery and two resistors $R_1=2\ \Omega$ and $R_2=4\ \Omega$ in series.

Required: Find current I using KVL.

Procedure:

1. Assume a current I flowing clockwise.
2. Traverse loop clockwise from bottom-left corner.
3. Pass through battery: - to + (Rise: +12V)
4. Pass through R_1 with current: (Drop: $-I \times 2$)
5. Pass through R_2 with current: (Drop: $-I \times 4$)
6. Return to start point = 0.

$$12 - 2I - 4I = 0$$

$$12 = 6I \Rightarrow I = 2\text{ A}$$

1.1.16 Circuit Classifications

In electrical engineering, circuits and elements are classified based on their V-I characteristics, energy contribution, and directional properties.

1. Linear vs. Non-linear

- **Linear Circuit:** A circuit whose parameters (R , L , C) do not change with voltage or current. The V-I characteristic is a straight line passing through the origin. **Ohm's law and Superposition principle apply.** (Example: Resistor).
- **Non-linear Circuit:** A circuit whose parameters change with voltage or current. The V-I characteristic is curved. (Example: Diode, Transistor, Incandescent bulb).

2. Bilateral vs. Unilateral

- **Bilateral Circuit:** Properties and characteristics are exactly the same in either direction of current flow. (Example: Resistor, Inductor).
- **Unilateral Circuit:** Properties change based on the direction of current flow. (Example: PN Junction Diode—conducts forward, blocks reverse).

3. Active vs. Passive

- **Active Element:** A component capable of generating or supplying electrical energy continuously. (Example: Voltage Source, Current Source, Operational Amplifier).
- **Passive Element:** A component that absorbs, dissipates, or temporarily stores energy, but cannot supply energy continuously. (Example: Resistor, Capacitor, Inductor).

Comparison Table

CATEGORY	LINEAR	NON-LINEAR	BILATERAL	UNILATERAL	ACTIVE	PASSIVE
V-I Graph	Straight line	Curve	Symmetrical in quadrants	Asymmetrical	N/A	N/A
Current Direction	Doesn't matter	Doesn't strictly matter	Doesn't matter	Changes behavior	Supplies power	Absorbs power
Examples	Standard Resistor	Diode	Resistor, AC line	Diode, SCR	Battery, Generator	Resistor, Coil

1.1.17 Section Summary

Key Equations Table

CONCEPT	EQUATION	NOTES
Series R	$R_{eq} = R_1 + R_2 + \dots$	Voltage divides
Parallel R	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	Current divides
Voltage Divider	$V_k = V_{total} \times \frac{R_k}{R_{eq}}$	Series only
Current Divider	$I_1 = I_{total} \times \frac{R_2}{R_1 + R_2}$	For 2 parallel branches
$\Delta \rightarrow Y$	$R_a = \frac{R_{ab}R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$	Denominator is sum of Δ
$Y \rightarrow \Delta$	$R_{ab} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c}$	Numerator is sum of pairs
KCL	$\sum I_{in} = \sum I_{out}$	Charge conservation
KVL	$\sum V = 0$	Energy conservation

Conceptual Questions

1. Why does the equivalent resistance of a parallel circuit always end up smaller than the smallest individual branch resistance?
 2. A diode behaves differently when the battery is reversed. Based on circuit classifications, how would you classify a diode?
 3. Explain why KCL is said to be based on the principle of conservation of charge.
 4. If one light bulb blows out in a series string, they all go out. Why doesn't this happen in house wiring?
 5. When transforming identical resistors from Star to Delta, does the resistance per branch increase or decrease, and by what factor?
-

Numerical Problems

Level 1

1. Two resistors of $12\ \Omega$ and $24\ \Omega$ are connected in parallel. Calculate the equivalent resistance.
2. Find the voltage drop across a $50\ \Omega$ resistor that is in series with a $150\ \Omega$ resistor, if the total applied voltage is 200V.

Level 2

3. A current of 10A flows into a parallel combination of $R_1=6\ \Omega$ and $R_2=4\ \Omega$. Use the current divider rule to find the current in R_1 .
4. Three resistors $10\ \Omega$, $10\ \Omega$, and $10\ \Omega$ are connected in Delta. They are to be replaced by an equivalent Star network. What is the value of each resistor in the Star network?

Level 3 (NEC Challenge)

5. A bridge circuit has a Delta loop with values $10\ \Omega$, $20\ \Omega$, and $30\ \Omega$. Convert it to Star. Then, place a $10\ \Omega$ resistor in series with the newly found R_a . What is the total resistance of that single branch now?
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NEC-Style MCQs

1. In a series circuit, which of the following remains constant across all elements?
 - a) Voltage
 - b) Current

- c) Power
 - d) Resistance
2. Three resistors of $30\ \Omega$ each are connected in delta. The equivalent star resistance per phase will be:
- a) $90\ \Omega$
 - b) $30\ \Omega$
 - c) $10\ \Omega$
 - d) $3\ \Omega$
3. Kirchhoff's Voltage Law (KVL) is based on the law of conservation of:
- a) Charge
 - b) Momentum
 - c) Mass
 - d) Energy
4. A network that contains at least one source of energy is called a(n):
- a) Passive network
 - b) Active network
 - c) Linear network
 - d) Bilateral network
5. The total resistance of two identical resistors in parallel is:
- a) Double of one resistor
 - b) Half of one resistor
 - c) Equal to one resistor
 - d) Zero
6. Which material property causes insulators to block current flow?
- a) They have overlapping valence and conduction bands
 - b) They have no tightly bound electrons
 - c) They have a very large energy gap and tightly bound valence electrons
 - d) They are very dense materials
7. If $V=100\text{V}$ is applied to $R_1=20\ \Omega$ and $R_2=30\ \Omega$ in series, what is the voltage across R_2 ?
- a) 40V
 - b) 60V
 - c) 50V
 - d) 100V
8. A circuit whose behavior depends upon the direction of current is known as:
- a) Unilateral
 - b) Bilateral

- c) Linear
- d) Active

9. Using KCL at a node, if currents 3A and 5A are entering, and I_x is leaving, what is I_x ?

- a) 2A
- b) -2A
- c) 8A
- d) 15A

10. Superposition theorem is applicable only to circuits that are:

- a) Non-linear
 - b) Linear
 - c) Unilateral
 - d) Active only
-

Answers

Conceptual Questions:

1. Adding paths in parallel provides additional routes for charge to flow, increasing overall conductance and thereby decreasing total resistance.
2. Unilateral and non-linear.
3. Charge cannot accumulate at a single point (node) over time in a stable circuit; thus, charge coming in must instantly equal charge going out.
4. House wiring is connected in parallel. Each appliance has its own independent loop with the voltage source.
5. The resistance increases by a factor of 3 ($R_\Delta = 3R_Y$).

Numerical Problems:

1. $R_{eq} = (12 \times 24) / (12 + 24) = 288 / 36 = 8 \, \Omega$.
2. $V_1 = 200 \times (50 / (50 + 150)) = 200 \times (50 / 200) = 50V$.
3. $I_1 = 10 \times (4 / (6 + 4)) = 10 \times (4 / 10) = 4A$.
4. $R_Y = 10 / 3 = 3.33 \, \Omega$.
5. From previous example, $R_a = 5 \, \Omega$. In series with $10 \, \Omega$, total is $15 \, \Omega$.

MCQs:

1. b, 2. c, 3. d, 4. b, 5. b, 6. c, 7. b, 8. a, 9. c, 10. b
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Section 1.2: Network Theorems and AC Circuits (AExE0102)

1. Introduction

This section forms the backbone of electrical circuit analysis. Understanding how to simplify complex networks using theorems like Thevenin's and Norton's, and how energy behaves in AC circuits with resistors, inductors, and capacitors, is fundamental to every branch of electrical and electronics engineering. The NEC Engineering License Exam heavily tests these concepts because they bridge pure theory with practical engineering design.

Part A: Network Theorems

1. Superposition Theorem

1.1 Statement and Conditions

The **Superposition Theorem** states that in any linear, active, bilateral network containing more than one independent source, the response (voltage or current) in any element is the algebraic sum of the responses caused by each independent source acting alone, while all other independent sources are deactivated.

Conditions for validity:

- The circuit must be **linear** (obeys Ohm's law, components don't change values with voltage/current).
- Applicable to voltage and current, but **NOT to power** (since power is proportional to the square of voltage/current, which is nonlinear).

1.2 Procedure

1. Identify all independent sources in the circuit.
2. Select one independent source to remain active.

3. **Deactivate** all other independent sources:
 - Voltage sources → **Short Circuit** (0V)
 - Current sources → **Open Circuit** (0A)
 - *Note: Dependent sources remain active.*
4. Calculate the desired voltage or current due to this single source.
5. Repeat steps 2-4 for every independent source.
6. Algebraically sum the contributions to find the total response.

1.3 Worked Numerical (Circuit with 2 Voltage Sources)

Given: A circuit with $V_1=12V$, $R_1=4\Omega$, $V_2=6V$, $R_2=2\Omega$, $R_L=4\Omega$ forming a T-network where R_L is the central shunt branch.

Required: Current through R_L (I_L) using superposition.

Step 1: Active $V_1=12V$, deactivate V_2 (Short Circuit)

The circuit becomes V_1 in series with R_1 , driving the parallel combination of R_L and R_2 .

Equivalent resistance: $R_{eq1}=R_1+(R_L||R_2)=4+(4||2)=4+1.33=5.33\Omega$

Total current $I_{T1}=12/5.33=2.25A$

Current through R_L (using current divider): $I_{L1}=I_{T1}\times\frac{R_2}{R_L+R_2}=2.25\times\frac{2}{4+2}=0.75A$ (downward)

Step 2: Active $V_2=6V$, deactivate V_1 (Short Circuit)

The circuit becomes V_2 in series with R_2 , driving the parallel combination of R_L and R_1 .

Equivalent resistance: $R_{eq2}=R_2+(R_L||R_1)=2+(4||4)=2+2=4\Omega$

Total current $I_{T2}=6/4=1.5A$

Current through R_L : $I_{L2}=I_{T2}\times\frac{R_1}{R_L+R_1}=1.5\times\frac{4}{4+4}=0.75A$ (downward)

Step 3: Algebraic Sum

$I_L=I_{L1}+I_{L2}=0.75+0.75=1.5A$

(downward)

Common Mistake: Trying to calculate total power by adding power contributions from individual sources. Total power $P \neq P_1 + P_2$.

2. Thevenin's Theorem

2.1 Statement

Any linear two-terminal network of independent sources, dependent sources, and resistors can be replaced by an equivalent circuit consisting of a single voltage source (V_{th}) in series with a single resistor (R_{th}).

2.2 Complete 10-Step Procedure

1. **Identify load:** Locate the load resistor R_L where the response is to be found.
2. **Remove load:** Disconnect R_L , leaving terminals open (say A and B).
3. **Calculate open-circuit voltage (V_{th}):** Find the voltage across A-B.
4. **Deactivate independent sources:** Voltage sources to short circuits, current sources to open circuits.
5. **Calculate equivalent resistance (R_{th}):** Find the resistance looking into terminals A-B.
6. **Construct Thevenin equivalent:** Draw V_{th} in series with R_{th} .
7. **Reconnect load:** Place R_L back across terminals A-B.
8. **Calculate load current:** $I_L = V_{th} / (R_{th} + R_L)$
9. **Calculate load voltage:** $V_L = I_L \times R_L$
10. **Calculate load power:** $P_L = I_L^2 \times R_L$

2.3 Worked Numerical (Level 2)

Given: Network with $V=10V$ series $R_1=2\Omega$, parallel $R_2=3\Omega$, series $R_3=1\Omega$, connected to $R_L=4\Omega$.

Required: Current in R_L using Thevenin.

1. Remove R_L .
2. V_{th} across A-B is voltage across R_2 . Voltage divider: $V_{th} = 10 \times 3 / (2 + 3) = 6V$.
3. Short 10V source. R_{th} is $R_3 + (R_1 \parallel R_2) = 1 + (2 \parallel 3) = 1 + 1.2 = 2.2\Omega$.
4. Thevenin equivalent: 6V in series with 2.2Ω .
5. Reconnect $R_L=4\Omega$. $I_L = 6 / (2.2 + 4) = 6 / 6.2 = 0.967A$.

3. Norton's Theorem

3.1 Statement

Any linear two-terminal network can be replaced by an equivalent circuit consisting of a single current source (I_N) in parallel with a single resistor (R_N).

3.2 Procedure

1. Identify and remove load R_L .
2. **Short-circuit** the load terminals.
3. Calculate the short-circuit current (I_N) flowing through the shorted terminals.
4. Calculate R_N (exactly the same procedure as R_{th}).
5. Construct Norton equivalent ($I_N || R_N$).
6. Reconnect R_L and calculate $I_L = I_N \times \frac{R_N}{R_N + R_L}$.

3.3 Relationship between Thevenin and Norton

- $R_N = R_{th}$
- $V_{th} = I_N \times R_N$
- $I_N = \frac{V_{th}}{R_{th}}$

This is an application of source transformation.

4. Maximum Power Transfer Theorem

4.1 Statement

Maximum power is transferred from a source to a load when the load resistance equals the Thevenin equivalent resistance of the source network as viewed from the load terminals ($R_L = R_{th}$).

4.2 Derivation

1. Power to load: $P_L = I_L^2 R_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 R_L$
2. To find maximum, take derivative with respect to R_L and equate to zero:

$$\frac{dP_L}{dR_L} = V_{th}^2 \left[\frac{1}{(R_{th} + R_L)^2} - \frac{R_L}{(R_{th} + R_L)^4} \right] = 0$$

3. Solving numerator gives: $R_{th} + R_L - 2R_L = 0 \Rightarrow R_L = R_{th}$

4. Substituting $R_L = R_{th}$ back into power equation:

$$P_{max} = V_{th}^2 / 4R_{th}$$

4.3 Efficiency

At maximum power transfer, $P_{load} = P_{loss} = I^2 R_{th}$. Total power generated is $2 \times P_{load}$.

$$\text{Efficiency } \eta = \frac{P_{load}}{P_{total}} \times 100\% = 50\%.$$

Engineering Intuition: 50% efficiency is terrible for a power grid (we want $\eta > 95\%$), but acceptable in telecommunications (like an antenna receiving a weak signal) where capturing the maximum possible power is more critical than efficiency.

Part B: R-L, R-C, R-L-C Circuits

5. R-L Circuit (AC)

- **Impedance:** $Z = R + jX_L = R + j\omega L$
- **Magnitude:** $|Z| = \sqrt{R^2 + X_L^2}$
- **Phase angle:** $\phi = \tan^{-1}(X_L/R)$
- **Current:** Lags voltage by angle ϕ .

6. R-C Circuit (AC)

- **Impedance:** $Z = R - jX_C = R - j1/\omega C$
- **Magnitude:** $|Z| = \sqrt{R^2 + X_C^2}$
- **Phase angle:** $\phi = \tan^{-1}(-X_C/R)$
- **Current:** Leads voltage by angle ϕ .

7. R-L-C Series Circuit (AC)

- **Impedance:** $Z = R + j(X_L - X_C)$
- **Three cases:**
 1. $X_L > X_C$: Inductive (current lags)

2. $X_L < X_C$: Capacitive (current leads)
3. $X_L = X_C$: Resonance (current in phase with voltage)

Part C: Resonance

8. Series Resonance

Occurs when inductive reactance equals capacitive reactance: $X_L = X_C$

- $\omega L = 1/\omega C \Rightarrow \omega^2 = 1/LC \Rightarrow f_0 = 1/(2\pi\sqrt{LC})$
- **Characteristics at resonance:**
- $Z = R$ (Minimum impedance)
- Current is maximum ($I = V/R$)
- Power factor is unity ($\text{pf} = 1$)
- **Quality Factor (Q):** Voltage magnification. $Q = X_L/R = 1/R\sqrt{LC}$
- **Bandwidth:** $\Delta f = f_0/Q = R/(2\pi L)$

9. Parallel Resonance

For a practical tank circuit (inductor with internal resistance R in parallel with a capacitor):

- Resonant frequency: $f_0 = 1/(2\pi\sqrt{LC - R^2L^2})$
- If R is very small, $f_0 \approx 1/(2\pi\sqrt{LC})$
- **At resonance:** Impedance is maximum, Current from source is minimum.

Part D: Active and Reactive Power

10. Active Power (P)

- $P = VI \cos(\phi)$ (Watts, W)
- Real power that does useful work (e.g., turns a motor, generates heat).

11. Reactive Power (Q)

- $Q = VI \sin(\phi)$ (Volt-Amperes Reactive, VAR)
- Power that oscillates back and forth between source and reactive elements (inductors/capacitors). Maintains magnetic/electric fields.

12. Apparent Power (S)

- $S=VI$ (Volt-Amperes, VA)
- $S=P+jQ$ (Complex power)
- Magnitude: $|S|=\sqrt{P^2+Q^2}$

13. Power Factor (pf)

- $\text{pf}=\cos(\phi)=P/S$
 - Indicates how effectively apparent power is converted to useful work.
-

End-of-Section

Conceptual Questions

1. Why does the Superposition Theorem fail for calculating power?
2. Explain the difference between Thevenin's and Norton's equivalent circuits.
3. Why is maximum power transfer not utilized in high-voltage power transmission?
4. What does a leading power factor imply about the circuit load?
5. How does bandwidth relate to the Quality factor in a resonant circuit?

Numerical Problems

Level 1: Find the resonant frequency of an RLC series circuit where $L=10\text{mH}$ and $C=1\mu\text{F}$.

Level 2: An AC circuit draws 10A at 220V with a pf of 0.8 lagging. Find Active, Reactive, and Apparent Power.

Level 3: Find the Thevenin equivalent of a circuit containing a 12V source in series with 5Ω , in parallel with a 2A current source.

NEC-Style MCQs

1. At series resonance, the circuit impedance is:
 - a) Maximum
 - b) Minimum
 - c) Zero
 - d) Infinite

Answer: b

Section 1.3: Alternating Current Fundamentals (AExE0103)

1. Principle of AC Generation

Introduction

An Alternating Current (AC) is a current whose magnitude changes with time and whose direction reverses periodically. Unlike Direct Current (DC), which flows constantly in one direction, AC is dynamic and forms the backbone of modern electrical power grids.

Why is it important?

AC is preferred over DC for power generation and transmission because AC voltage can be easily stepped up or stepped down using transformers. This allows power to be transmitted at very high voltages (reducing I^2R line losses) and distributed at safer, lower voltages.

Basic Concept

AC is generated using an alternator (AC generator). The fundamental principle relies on Faraday's Law of Electromagnetic Induction: when a conductor rotates within a uniform magnetic field (or when a magnetic field rotates around a stationary conductor), an electromotive force (EMF) is induced in the conductor.

Physical Meaning

Imagine a rectangular wire coil rotating at a constant angular velocity inside a constant magnetic field. As it turns, the area of the coil perpendicular to the magnetic field changes, altering the magnetic flux passing through it. This changing flux induces a voltage.

The Rotating Coil Diagram

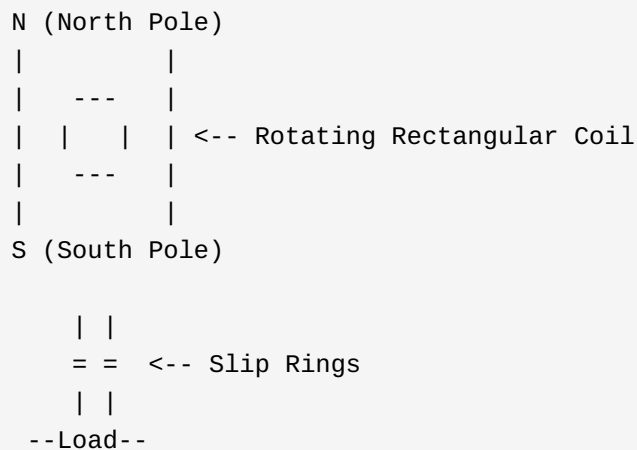


Figure: Basic AC Generator with a rotating coil, magnetic field, and slip rings.

Mathematical Formulation and Derivation

Let a coil of N turns and area A rotate at a constant angular velocity ω in a uniform magnetic field of flux density B .

The magnetic flux Φ linked with the coil at any time t is:

$$\Phi = B \cdot A \cdot \cos(\theta) = B \cdot A \cdot \cos(\omega t)$$

According to Faraday's Law, the induced EMF (e) is the negative rate of change of flux linkage:

$$e = -N \frac{d\Phi}{dt}$$

$$e = -N \frac{d}{dt}(BA \cos(\omega t))$$

$$e = -NBA(-\omega \sin(\omega t))$$

$$e = NBA\omega \sin(\omega t)$$

Let the maximum possible EMF (when $\sin(\omega t) = 1$) be $V_m = NBA\omega$. Then:

$$v(t) = V_m \sin(\omega t)$$

This shows why the output voltage is perfectly **sinusoidal**. One complete mechanical rotation (360° or 2π radians) of a 2-pole machine corresponds to one complete electrical cycle of the AC waveform.

2. Equations of AC Voltage and Current

The general equations for sinusoidal AC quantities are:

Voltage:

$$v(t) = V_m \sin(\omega t + \phi)$$

Current:

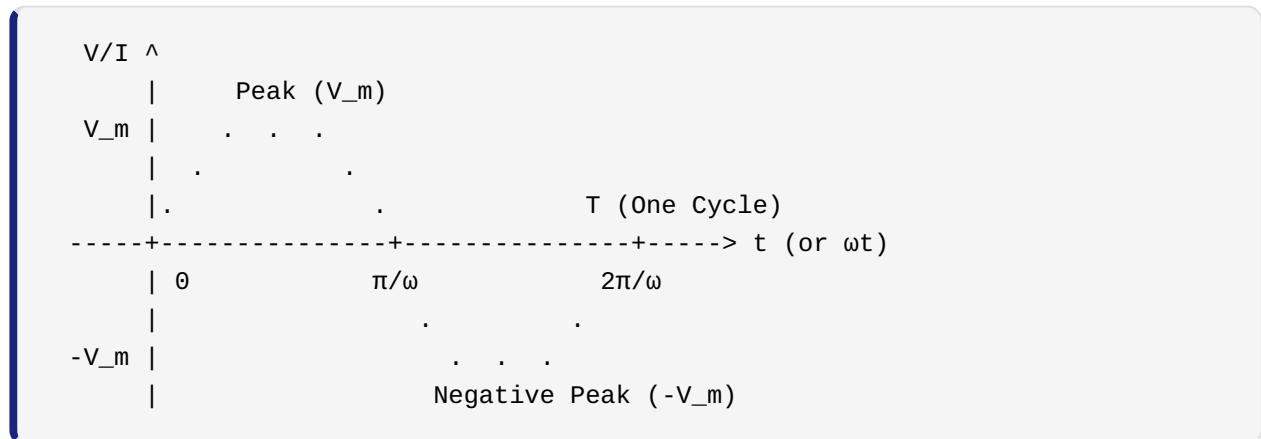
$$i(t) = I_m \sin(\omega t + \phi)$$

Variable Table

SYMBOL	MEANING	SI UNIT
$v(t), i(t)$	Instantaneous voltage/current at time t	Volts (V), Amperes (A)
V_m, I_m	Peak (maximum) voltage/current	Volts (V), Amperes (A)
ω	Angular frequency ($\omega = 2\pi f$)	Radians per second (rad/s)
f	Frequency (number of cycles per second)	Hertz (Hz)
T	Time period of one cycle ($T = 1/f$)	Seconds (s)
ϕ	Phase angle (initial angle at $t=0$)	Radians (rad) or Degrees (°)

3. AC Waveforms

Sinusoidal Waveform Diagram



- **Zero Crossings:** Occur at $0, \pi, 2\pi, \dots$ radians. The instantaneous value is zero.
- **Positive Peak:** Occurs at $\pi/2$ radians.
- **Negative Peak:** Occurs at $3\pi/2$ radians.
- **Standard Frequencies:** In Nepal, the standard grid frequency is **50 Hz**.
- Time period $T = 1/f = 1/50 = 0.02$ seconds (20 ms).

4. Peak Value (Maximum Value)

Definition: The maximum amplitude reached by an alternating quantity during one cycle is called its peak value (V_m or I_m).

- **Peak-to-Peak Value (V_{pp}):** The total voltage swing from the positive peak to the negative peak.

$$V_{pp} = 2V_m$$

5. Average Value

Definition: The arithmetic average of all instantaneous values over one half-cycle. (For a full pure sine wave, the average value is exactly zero, so we always define it over a half-cycle).

Derivation:

$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} v(\theta) d\theta$$

$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} V_m \sin(\theta) d\theta$$

$$V_{avg} = \frac{V_m}{\pi} [-\cos(\theta)]_0^{\pi}$$

$$V_{avg} = \frac{V_m}{\pi} (-\cos(\pi) + \cos(0))$$

$$V_{avg} = \frac{V_m}{\pi} (1+1) = \frac{2V_m}{\pi} \approx 0.637V_m$$

6. RMS Value (Root Mean Square)

Definition: The RMS value of an AC current is the equivalent DC current that would produce the same amount of heat in a given resistor over a given time. It is the most important measurement in AC engineering.

Derivation:

$$V_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} [v(\theta)]^2 d\theta}$$

$$V_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} (V_m \sin \theta)^2 d\theta}$$

Using the identity $\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$:

$$V_{rms} = V_m \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (1 - \cos(2\theta)) d\theta}$$

$$V_{rms} = V_m \sqrt{\frac{1}{2\pi} [\theta - \sin(2\theta)]_0^{2\pi}}$$

$$V_{rms} = V_m \sqrt{\frac{1}{2\pi} (2\pi - 0)} = V_m \sqrt{\frac{1}{2}} = \frac{V_m}{\sqrt{2}}$$

$$V_{rms} \approx 0.707 V_m$$

💡 **Engineering Intuition:** When you say the household supply in Nepal is "220V", that is the **RMS value**.

The peak voltage you are exposed to is $V_m = 220 \times \sqrt{2} \approx 311.1$ V!

7. Form Factor and Crest Factor

These factors describe the shape of the AC wave.

- **Form Factor (Kf):** Ratio of RMS value to Average value.

$$K_f = \frac{V_{rms}}{V_{avg}} = \frac{V_m/\sqrt{2}}{V_m/\pi} = \frac{\pi}{\sqrt{2}} \approx 1.11$$

- **Crest Factor (Kc or Peak Factor):** Ratio of Peak value to RMS value.

$$K_c = \frac{V_m}{V_{rms}} = \frac{V_m}{V_m/\sqrt{2}} = \sqrt{2} \approx 1.414$$

8. Relationships Summary Table

QUANTITY	FORMULA	VALUE FOR SINE WAVE
Peak Value	V_m	V_m
Average Value	$\frac{1}{\pi} \int_0^{\pi} v d\theta$	$0.637V_m$
RMS Value	$\sqrt{\frac{1}{2\pi} \int_0^{2\pi} v^2 d\theta}$	$0.707V_m$
Form Factor	RMS / Average	1.11
Crest Factor	Peak / RMS	1.414

9. Phase and Phase Difference

- **Phase Angle (ϕ):** Represents the initial state of the waveform at $t=0$.
- **Phase Difference:** The angular displacement between two AC quantities of the same frequency.
- If $v = V_m \sin(\omega t)$ and $i = I_m \sin(\omega t - \phi)$, the current **lags** the voltage by ϕ .
- If $v = V_m \sin(\omega t)$ and $i = I_m \sin(\omega t + \phi)$, the current **leads** the voltage by ϕ .
- **In-phase:** $\phi = 0^\circ$
- **Quadrature:** $\phi = 90^\circ$
- **Anti-phase:** $\phi = 180^\circ$

10. Three-Phase System

Need for Three-Phase

Single-phase power drops to zero twice per cycle, causing pulsating power. Three-phase power delivers a constant, non-pulsating total power to balanced loads, which reduces vibrations in motors and improves efficiency. It also requires less conductor material to transmit the same power compared to single-phase.

Generation

A three-phase alternator has three sets of coils arranged 120° apart mechanically. This produces three sinusoidal voltages of equal magnitude and frequency, but phase-shifted by 120° (electrical).

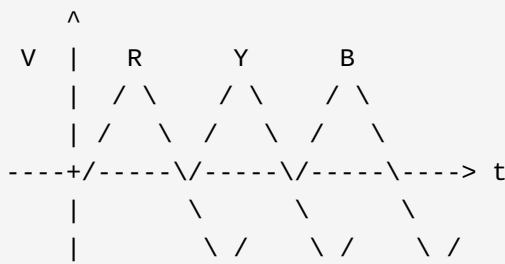
Phase Sequence: Typically Red, Yellow, Blue (R-Y-B).

$$v_R = V_m \sin(\omega t)$$

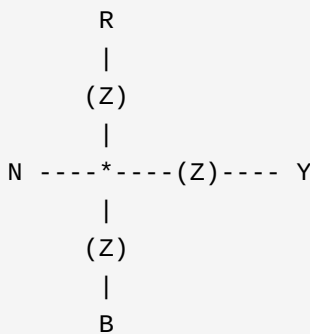
$$v_Y = V_m \sin(\omega t - 120^\circ)$$

$$v_B = V_m \sin(\omega t - 240^\circ) = V_m \sin(\omega t + 120^\circ)$$

Three-phase Waveforms:



Star (Y) Connection



- **Neutral wire** is available from the common star point (N).
- **Line Current (IL) vs Phase Current (Iph):** They are the same series path.

$$I_L = I_{ph}$$

- **Line Voltage (VL) vs Phase Voltage (Vph):**

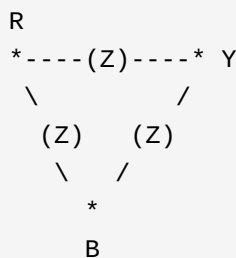
$$V_{RY} = V_{\rightarrow RN} - V_{\rightarrow YN}$$

Using phasor subtraction of two vectors separated by 120° :

$$V_L = \sqrt{V_{ph}^2 + V_{ph}^2 - 2V_{ph}V_{ph}\cos(120^\circ)} = 3V_{ph}$$

$$V_L = 3V_{ph}$$

Delta (Δ) Connection



- No neutral wire exists.
- **Line Voltage vs Phase Voltage:** The line terminals are directly connected across the phase coils.

$$V_L = V_{ph}$$

- **Line Current vs Phase Current:**

$$I_{\rightarrow R} = I_{\rightarrow RY} - I_{\rightarrow BR}$$

By phasor subtraction:

$$I_L = 3I_{ph}$$

Three-Phase Power

For a balanced system, total power is the sum of power in 3 phases.

$$P_{total} = 3 \times P_{phase} = 3 \times V_{ph} \times I_{ph} \times \cos(\phi)$$

Expressing in terms of Line quantities:

- **In Star:** $V_{ph} = V_L/3$, $I_{ph} = I_L$.

$$P = 3 \times (V_L/3) \times I_L \times \cos(\phi) = 3V_L I_L \cos(\phi)$$

- **In Delta:** $V_{ph} = V_L$, $I_{ph} = I_L/3$.

$$P = 3 \times V_L \times (I_L/3) \times \cos(\phi) = 3V_L I_L \cos(\phi)$$

- **Active Power:** $P = 3V_L I_L \cos(\phi)$ (Watts, W)
- **Reactive Power:** $Q = 3V_L I_L \sin(\phi)$ (Volt-Amperes Reactive, VAR)
- **Apparent Power:** $S = 3V_L I_L$ (Volt-Amperes, VA)

Star vs Delta Comparison Table

FEATURE	STAR (Y) CONNECTION	DELTA (Δ) CONNECTION
Neutral terminal	Present	Absent
Line vs Phase Voltage	$V_L = 3V_{ph}$	$V_L = V_{ph}$
Line vs Phase Current	$I_L = I_{ph}$	$I_L = 3I_{ph}$
Application	Distribution, loads requiring neutral	Transmission, large motors

11. Section Summary

Worked Numerical Problem - LEVEL 2

Problem: A 400 V (line-to-line), 50 Hz, three-phase supply is connected to a balanced star-connected load. The phase current is 10 A, lagging the phase voltage by 30° . Calculate the phase voltage, total active power, and total apparent power.

Given:

- Line Voltage $V_L = 400$ V
- Frequency $f = 50$ Hz
- Phase Current $I_{ph} = 10$ A
- Phase Angle $\phi = 30^\circ$ (lagging)
- Connection = Star (Y)

Required: V_{ph} , P_{total} , S_{total}

Principle & Formulas:

For Star: $V_{ph} = V_L/3$, $I_L = I_{ph}$

$$P = 3V_L I_L \cos(\phi) \quad S = 3V_L I_L$$

Substitution & Calculation:

1. $V_{ph} = 400/3 = 230.94$ V
2. $I_L = I_{ph} = 10$ A
3. $P = 3 \times 400 \times 10 \times \cos(30^\circ) = 3 \times 4000 \times (3/2) = 6000$ W = 6 kW
4. $S = 3 \times 400 \times 10 = 6928.2$ VA = 6.93 kVA

Final Answer:

- Phase Voltage = 230.9 V
- Active Power = 6.0 kW
- Apparent Power = 6.93 kVA

Engineering Interpretation: A typical low-voltage distribution in Nepal provides 400V between lines, which naturally gives $400/3 \approx 230$ V for single-phase household loads across line and neutral.

Common Mistake: Using phase voltage instead of line voltage in the $3VI$ power formula, or using line voltage in the $3VI$ formula. Remember: $P = 3V_L I_L \cos\phi = 3V_{ph} I_{ph} \cos\phi$.

5 Conceptual Questions

1. Why does a pure sine wave have an average value of zero over a full cycle?
2. Explain the physical significance of the RMS value. Why is it more useful than the peak value?
3. In a rotating coil AC generator, what determines the frequency of the generated voltage?
4. Why is a three-phase system preferred over a single-phase system for power transmission?
5. Under what conditions is the neutral current zero in a 3-phase, 4-wire star-connected system?

5 Numerical Problems

Level 1

1. An AC voltage is given by $v(t) = 311\sin(314t)$. Find the peak voltage, RMS voltage, and frequency.
2. Calculate the form factor and crest factor of a perfectly square wave (where $V_m = V_{rms} = V_{avg}$).

Level 2

3. Three identical impedances of $10\angle 30^\circ \Omega$ are connected in Delta across a 400V, 3-phase supply. Find the phase current, line current, and total active power.
4. An AC current has an RMS value of 15 A and a frequency of 60 Hz. Write its time-domain equation $i(t)$, assuming zero phase angle.

Level 3

5. A factory has a balanced 3-phase load drawing 50 kW at 0.8 power factor lagging from a 415 V supply. If the load is star-connected, determine the phase voltage, line current, and impedance per phase.

10 NEC-Style MCQs

1. The form factor of a standard sinusoidal AC wave is:
A) 0.707
B) 1.11
C) 1.414
D) 0.637

-
2. In Nepal, the standard domestic AC supply frequency is:
- A) 60 Hz
 - B) 50 Hz
 - C) 100 Hz
 - D) 120 Hz
3. The RMS value of an AC wave is equivalent to a DC value that produces the same:
- A) Voltage drop
 - B) Magnetic field
 - C) Heating effect
 - D) Charge transfer
4. In a pure sinusoidal wave, the relationship between peak value (V_m) and RMS value (V_{rms}) is:
- A) $V_{rms} = V_m/2$
 - B) $V_{rms} = V_m/\sqrt{2}$
 - C) $V_{rms} = 2V_m$
 - D) $V_{rms} = V_m/\pi$
5. Which connection requires a neutral wire for supplying unbalanced single-phase loads?
- A) Delta connection
 - B) Star connection
 - C) Series connection
 - D) Parallel connection
6. In a balanced 3-phase Delta (Δ) connection:
- A) $V_L = 3V_{ph}$ and $I_L = I_{ph}$
 - B) $V_L = V_{ph}$ and $I_L = 3I_{ph}$
 - C) $V_L = 3V_{ph}$ and $I_L = 3I_{ph}$
 - D) $V_L = V_{ph}$ and $I_L = I_{ph}$
7. The total active power in a balanced 3-phase system is given by:
- A) $3V_L I_L \cos\phi$
 - B) $3V_L I_L \cos\phi$
 - C) $3V_{ph} I_{ph} \cos\phi$
 - D) $V_L I_L \cos\phi$
8. If two alternating quantities A and B are represented by $A = A_m \sin(\omega t)$ and $B = B_m \sin(\omega t + 90^\circ)$, then:
- A) A leads B by 90°
 - B) B leads A by 90°

- C) A and B are in phase
- D) A lags B by 180°

9. A single-phase AC voltage is $v(t)=141.4\sin(314t)$. Its approximate RMS value is:

- A) 100 V
- B) 141.4 V
- C) 200 V
- D) 314 V

10. Why are three-phase systems generally preferred over single-phase for transmission?

- A) They require exactly three wires, saving cost
- B) They produce constant power instead of pulsating power
- C) They generate less heat in the generator
- D) They do not require transformers

MCQ Answers

- 1. B
 - 2. B
 - 3. C
 - 4. B
 - 5. B
 - 6. B
 - 7. A
 - 8. B
 - 9. A
 - 10. B
-

Section 1.4: Semiconductor Devices (AExE0104)

Syllabus Coverage: Semiconductor diode and its characteristics, BJT Configuration and biasing, small and large signal model, working principle and application of MOSFET and CMOS.

Introduction

Semiconductor devices form the fundamental building blocks of modern electronics. From a simple diode rectifying AC power to billions of CMOS transistors inside a microprocessor, understanding how semiconductors behave under various electrical conditions is critical for any electrical or electronics engineer. This section covers the fundamental physics of semiconductors, diodes, BJTs, MOSFETs, and CMOS technology.

Part A: Semiconductor Fundamentals (Prerequisite)

1. Conductors, Semiconductors, Insulators

The electrical properties of materials are defined by their atomic structure, specifically their energy bands.

- **Valence Band:** The highest energy band completely filled with electrons at absolute zero temperature.
- **Conduction Band:** The energy band above the valence band. Electrons here are free to move and conduct electricity.
- **Forbidden Gap (E_g):** The energy difference between the conduction and valence bands where no electron states can exist.

PROPERTY	CONDUCTORS	SEMICONDUCTORS	INSULATORS
Band Gap (E_g)	Overlapping (E _g ≈0)	Small (E _g ≈1 eV)	Large (E _g >5 eV)
Conductivity	Very High	Moderate, controllable	Negligible
Examples	Copper, Gold, Aluminum	Silicon, Germanium	Glass, Rubber, Diamond

 **Engineering Intuition:** Semiconductors are special not because they conduct "semi-well", but because their conductivity is *controllable* via temperature, voltage, or doping.

2. Intrinsic Semiconductor

An intrinsic semiconductor is a perfectly pure semiconductor crystal (e.g., pure Silicon or Germanium).

- * **Covalent Bonding:** Each atom shares 4 valence electrons with its neighbors.
- * **Electron-Hole Pairs:** At room temperature (300K), thermal energy breaks some covalent bonds, creating free electrons and leaving behind vacancies called **holes** (which act as positive charge carriers).
- * **Carrier Concentration:** In intrinsic semiconductors, the number of electrons (n) equals the number of holes (p).

$$n=p=n_i$$

where n_i is the intrinsic carrier concentration.

3. Extrinsic Semiconductor

To make semiconductors useful, we deliberately introduce impurities in a process called **doping**.

FEATURE	N-TYPE SEMICONDUCTOR	P-TYPE SEMICONDUCTOR
Dopant Impurity	Pentavalent (Group V: P, As, Sb)	Trivalent (Group III: B, Al, Ga)
Dopant Name	Donor (donates an electron)	Acceptor (accepts an electron)
Majority Carriers	Electrons	Holes
Minority Carriers	Holes	Electrons
Fermi Level	Shifts towards Conduction Band	Shifts towards Valence Band

4. PN Junction

When P-type and N-type materials are joined, a **PN junction** is formed.

Formation of Depletion Region:

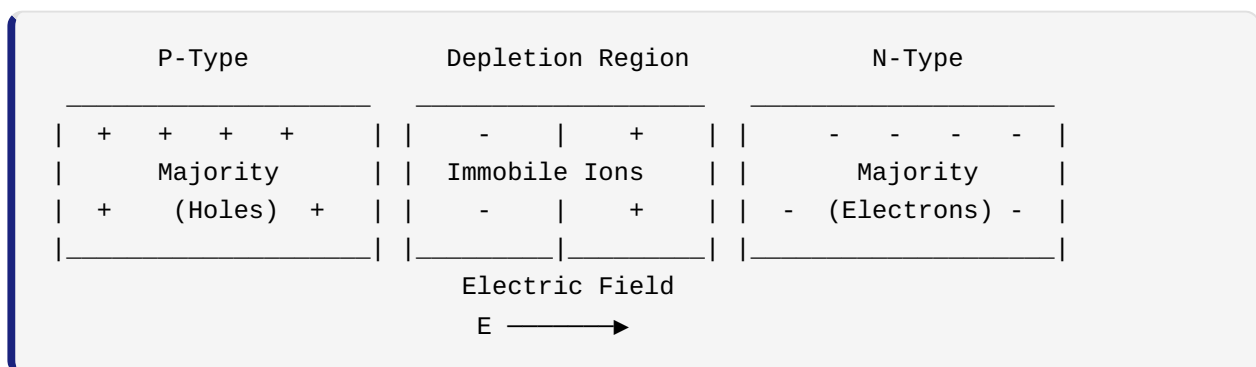
1. High concentration of holes in P and electrons in N causes them to diffuse across the junction.
2. As electrons cross to the P-side, they leave behind positively charged donor ions. Holes crossing to the N-side leave behind negatively charged acceptor ions.
3. These immobile ions create a **depletion region** (devoid of mobile charge carriers) and an internal electric field.

Built-in Potential (V_{bi}):

The electric field creates a barrier potential that opposes further diffusion.

* Silicon (Si): ≈ 0.7 V

* Germanium (Ge): ≈ 0.3 V



Part B: Semiconductor Diode

5. Diode Construction and Operation

A diode is simply a PN junction with electrical terminals attached. It allows current to flow in one direction but blocks it in the other.

* **Symbol:**

text Anode (A) ———▶|———— Cathode (K) (P-type) (N-type)

6. Forward Bias

When the positive terminal of a battery is connected to the Anode (P) and the negative to the Cathode (N):

- * The external voltage *opposes* the built-in potential.
- * The depletion region narrows.
- * Once $V_{\text{external}} > V_{\text{barrier}}$, large current flows.

* **Mathematical Formulation (Shockley Diode Equation):**

$$I = I_s(e^{V/V_T} - 1)$$

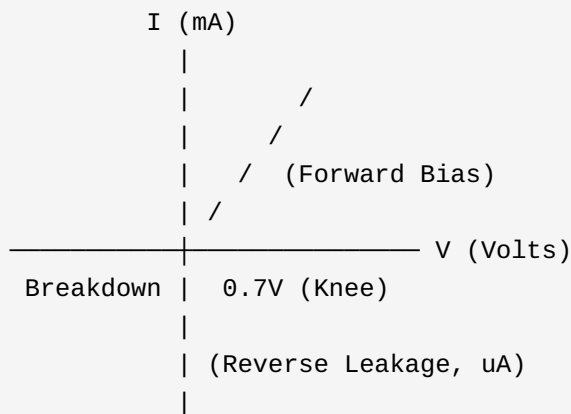
Symbol	Meaning	SI Unit
I	Diode current	A
I_s	Reverse saturation current	A
V	Voltage across diode	V
n	Ideality factor (1 to 2)	Unitless
V_T	Thermal voltage ($kT/q \approx 26\text{mV}$ at 300K)	V

7. Reverse Bias

When positive is connected to Cathode (N) and negative to Anode (P):

- * The external voltage *adds* to the built-in potential.
- * The depletion region widens.
- * Majority carrier current is blocked. Only a tiny minority carrier drift current (I_s , leakage current) flows.

8. V-I Characteristics



- **Knee/Cut-in Voltage:** The forward voltage at which current starts to increase rapidly (0.7V for Si).
- **Breakdown Voltage:** The reverse voltage at which the diode fails and conducts heavily in reverse.

9. Diode Models

1. **Ideal Diode Model:** Acts as a perfect switch. (0V drop when forward, 0A when reverse).
2. **Constant Voltage Drop (CVD) Model:** Forward-biased diode acts like a constant 0.7V battery opposing current flow.

Worked Numerical:

Given: A 10V DC source is in series with a Si diode and a 1 k Ω resistor.

Required: Find the current using the CVD model.

Principle: KVL.

Calculation: $10V - 0.7V - I \cdot 1000\Omega = 0 \Rightarrow I = 9.3V / 1000\Omega = 9.3 \text{ mA}$.

10. Breakdown

- **Zener Breakdown:** Occurs in heavily doped diodes. The depletion region is so thin that the strong electric field tears electrons from covalent bonds (tunneling). Usually occurs <5V.
- **Avalanche Breakdown:** Occurs in lightly doped diodes. High reverse voltage accelerates minority carriers, causing them to collide with atoms and free other electrons in an avalanche effect. Usually occurs >5V.

11. Diode Applications

- **Rectification:** Converting AC to DC (Half-wave, Full-wave).
- **Voltage Regulation:** Zener diodes in reverse breakdown maintain constant voltage.
- **Clipping/Clamping:** Limiting waveforms or shifting DC levels.

Part C: Bipolar Junction Transistor (BJT)

12. BJT Construction

A BJT consists of three alternating layers of P and N type semiconductors.

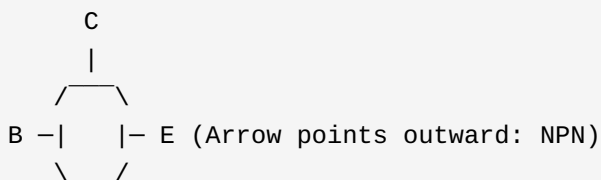
* **NPN:** Emitter (N), Base (P), Collector (N)

* **PNP:** Emitter (P), Base (N), Collector (P)

Doping and Sizing:

1. **Emitter (E):** Heavily doped (to inject many carriers).
2. **Base (B):** Very thin and lightly doped (so carriers pass through without recombining).
3. **Collector (C):** Moderately doped, largest area (to dissipate heat).

NPN Transistor Symbol



13. Transistor Action (For NPN)

To use a BJT as an amplifier, it operates in the **Active Region**:

1. **Base-Emitter (BE) Junction:** Forward-biased. Electrons are injected from the heavily doped emitter into the base.
2. **Base-Collector (BC) Junction:** Reverse-biased.
3. Because the base is very thin and lightly doped, 95-99% of electrons shoot straight through the base and are swept into the collector by the strong reverse-bias electric field.
4. Fundamental KCL Equation: $I_E = I_B + I_C$

14. Current Relationships

- **α (Common-Base Current Gain):** $\alpha = I_{CIE}$. Usually 0.95 to 0.998.
- **β (Common-Emitter Current Gain):** $\beta = I_{CIB}$. Usually 50 to 300.

Derivation of Relationship:

1. Start with $I_E = I_B + I_C$.
2. Divide by I_C : $I_E I_C = I_B I_C + 1$.
3. Substitute $1\alpha = 1\beta + 1$.
4. Rearrange: $\beta = \alpha 1 - \alpha$ and $\alpha = \beta \beta + 1$.

15. BJT Configurations

CHARACTERISTIC	COMMON EMITTER (CE)	COMMON BASE (CB)	COMMON COLLECTOR (CC)
Input/Output	Base / Collector	Emitter / Collector	Base / Emitter
Voltage Gain (A_v)	High	High	≈ 1
Current Gain (A_i)	High (β)	< 1 (α)	High ($\approx \beta$)
Power Gain	Very High	Moderate	Moderate
Input Impedance	Moderate ($\approx 1k\Omega$)	Very Low ($\approx 20\Omega$)	Very High (several $k\Omega$)
Output Impedance	Moderate	Very High	Low
Phase Shift	180° (Inverting)	0°	0°
Main Application	General amplification	High-frequency	Buffer (Impedance matching)

16. Operating Regions

REGION	BE JUNCTION	BC JUNCTION	APPLICATION
Cutoff	Reverse	Reverse	Open Switch (Logic 0)
Active	Forward	Reverse	Amplifier
Saturation	Forward	Forward	Closed Switch (Logic 1)

⚠ NEC Exam Trap: BJT Saturation means both junctions are forward-biased, VCE drops to $\approx 0.2V$, and IC is determined by the external circuit, NOT βI_B . Do not use $I_C = \beta I_B$ in saturation!

17. Biasing and Q-Point

Biasing sets the steady DC operating point (Q-point) in the middle of the active region so that AC signals can swing without clipping.

Voltage Divider Bias (Most Stable):

Uses resistors R1 and R2 to divide VCC and set a stable VB.

$$V_B \approx V_{CC} \frac{R_2}{R_1 + R_2}$$

(assuming

β

is large).

$$V_E = V_B - 0.7V$$

.

$$I_E = V_E / R_E \approx I_C$$

.

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

.

18. Small-Signal Model

Small signal analysis treats the BJT as a linear two-port network for small AC variations around the Q-point.

* **Transconductance (g_m):** The ratio of small change in output current to small change in input voltage.

$g_m = I_{CQ}/V_T$ where $V_T \approx 26\text{mV}$.

* **Input Resistance (r_{π}):** $r_{\pi} = \beta/g_m$.

* **Voltage Gain (A_v) of CE Amplifier:** $A_v \approx -g_m R_{C(\text{eff})}$.

19. Large-Signal Model vs Small-Signal

- **Large-Signal (Ebers-Moll):** Non-linear model covering all regions (cutoff, active, sat). Used for DC biasing and switching.
- **Small-Signal (Hybrid- π):** Linearized model. Valid only for small AC signals around a fixed DC point.

Part D: MOSFET

Metal-Oxide-Semiconductor Field-Effect Transistor

20. MOSFET Construction

Instead of a PN junction controlling the current (like BJT), a MOSFET uses an electric field across a dielectric (oxide) to control channel conductivity.

* **Terminals:** Gate (G), Source (S), Drain (D), Body/Substrate (B). (Body usually tied to Source).

* **Structure:** Metal Gate, separated from a semiconductor body by a thin layer of Silicon Dioxide (SiO_2).

21. Working Principle

- **Enhancement Mode (E-MOSFET):** Normally OFF. A Gate-Source voltage (V_{GS}) greater than a threshold (V_{th}) must be applied to "enhance" or create a conducting channel by attracting electrons to the surface under the oxide.
- **Depletion Mode (D-MOSFET):** Normally ON. Channel physically exists; varying V_{GS} modulates its depth.

22. Regions of Operation (Enhancement NMOS)

1. **Cutoff:** $V_{GS} < V_{th}$. No channel. $I_D = 0$.
2. **Linear / Triode:** $V_{GS} > V_{th}$ and $V_{DS} < V_{GS} - V_{th}$. Channel behaves like a voltage-controlled resistor.

$$I_D = k_n [(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}]$$
3. **Saturation:** $V_{GS} > V_{th}$ and $V_{DS} \geq V_{GS} - V_{th}$. Channel "pinches off". Current becomes independent of V_{DS} and acts as a constant current source.

$$I_D = \frac{k_n}{2} (V_{GS} - V_{th})^2$$

(Where $k_n = \mu_n C_{ox} W/L$ is the conduction parameter)

⚠ NEC Exam Trap: Notice that "Saturation" in a MOSFET means it is acting as a constant current source (ideal for amplification). In a BJT, "Saturation" means it acts as a short circuit (switch ON). The terminology is exactly opposite in meaning for amplifiers!

23. MOSFET Applications

- Microprocessors, memory chips (Digital logic).
- Power electronics (Power MOSFETs for high current switching).
- Analog amplification (due to high input impedance).

24. BJT vs MOSFET Comparison

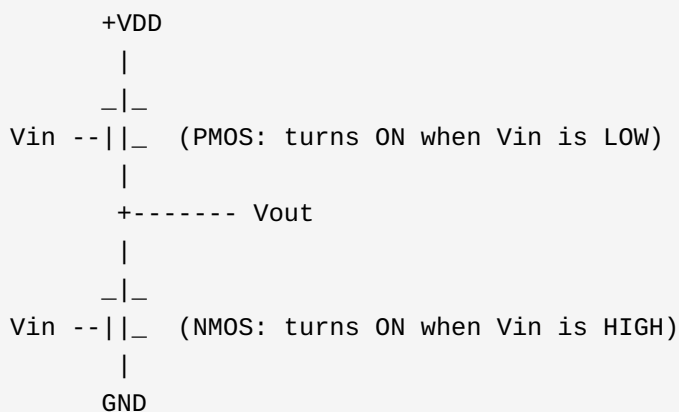
FEATURE	BJT	MOSFET
Control Variable	Current controlled (I_B controls I_C)	Voltage controlled (V_{GS} controls I_D)
Carriers	Bipolar (electrons and holes)	Unipolar (only majority carriers)
Input Impedance	Low to Moderate	Extremely High (virtually infinite at DC)
Size / Packing	Larger	Very small (excellent for VLSI)
Thermal Stability	Poor (Prone to thermal runaway)	Excellent (Positive temp coefficient)

Part E: CMOS

25. CMOS Concept

Complementary MOS (CMOS) uses both NMOS and PMOS transistors in pairs.

CMOS Inverter:



- When V_{in} =HIGH (VDD): NMOS is ON, PMOS is OFF. V_{out} is pulled to GND .
- When V_{in} =LOW (GND): NMOS is OFF, PMOS is ON. V_{out} is pulled to VDD .

26. CMOS Advantages

1. **Zero Static Power:** In steady state (either Logic 0 or 1), one transistor is always OFF. There is no direct path from VDD to GND . Power is only consumed during *switching* (dynamic power).
2. **High Noise Margin:** VTC is nearly ideal.
3. **High Density:** Tiny footprint, making billions of transistors on a chip possible.

27. Applications

Almost all modern digital ICs, microprocessors, ASICs, and memory (SRAM) are built using CMOS technology.

End-of-Section

Key Equations

CONCEPT	EQUATION
Intrinsic Carriers	$n=p=n_i$
Shockley Diode	$I=I_s(e^{V/V_T}-1)$
BJT Current	$I_E=I_B+I_C$
BJT Alpha/Beta	$\beta=\alpha/(1-\alpha); \alpha=\beta/(\beta+1)$
MOSFET Saturation	$I_D=k_n/2(V_{GS}-V_{th})^2$
BJT Transconductance	$g_m=I_C/V_T$

5 Conceptual Questions

1. Why does an intrinsic semiconductor behave like an insulator at absolute zero temperature (0 K)?
2. Why is the base of a BJT made very thin and lightly doped?
3. Explain why the input impedance of a MOSFET is vastly higher than that of a BJT.
4. Why does a BJT enter thermal runaway while a MOSFET does not?
5. Why is CMOS the dominant technology for digital integrated circuits?

5 Worked Numerical Problems

LEVEL 1 (Foundation)

Q1: A silicon diode has a reverse saturation current of 10 nA at 300K. Calculate the forward current if a forward voltage of 0.6 V is applied. (Assume $n=1$, $V_T=26\text{mV}$)

* **Given:** $I_s=10\text{ nA}=10\times 10^{-9}\text{A}$, $V=0.6\text{V}$, $V_T=0.026\text{V}$.

* **Formula:** $I=I_s e^{V/V_T}$

* **Calculation:** $I=(10\times 10^{-9})\times e^{0.6/0.026}=10\times 10^{-9}\times 1.05\times 10^{10}=0.105\text{ A}=105\text{ mA}$.

* **Answer:** 105 mA.

Q2: A BJT has an α of 0.99. If the base current is $20\text{ }\mu\text{A}$, calculate the collector current and emitter current.

* **Given:** $\alpha=0.99$, $I_B=20\text{ }\mu\text{A}$.

* **Formula:** $\beta = \alpha / (1 - \alpha)$, $I_C = \beta I_B$, $I_E = I_C + I_B$.

* **Calculation:** $\beta = 0.99 / 0.01 = 99$. $I_C = 99 \times 20 \mu A = 1.98 \text{ mA}$. $I_E = 1.98 \text{ mA} + 0.02 \text{ mA} = 2.0 \text{ mA}$.

* **Answer:** $I_C = 1.98 \text{ mA}$, $I_E = 2.0 \text{ mA}$.

LEVEL 2 (Engineering)

Q3: For a CE amplifier, $V_{CC} = 12V$, $R_C = 2k\Omega$, and $\beta = 100$. If it is biased such that $V_{CE} = 6V$ (midpoint), calculate the required base current I_B . (Assume $R_E = 0$).

* **Given:** $V_{CC} = 12V$, $V_{CE} = 6V$, $R_C = 2000\Omega$.

* **Principle:** KVL at output: $V_{CC} - I_C R_C - V_{CE} = 0$.

* **Calculation:** $I_C = (12 - 6) / 2000 = 3 \text{ mA}$. $I_B = I_C / \beta = 3 \text{ mA} / 100 = 30 \mu A$.

* **Answer:** $30 \mu A$.

Q4: An NMOS transistor has $V_{th} = 1V$ and $k_n = 1 \text{ mA/V}^2$. It is biased with $V_{GS} = 3V$ and $V_{DS} = 4V$. Determine the region of operation and the drain current.

* **Given:** $V_{th} = 1V$, $V_{GS} = 3V$, $V_{DS} = 4V$.

* **Principle:** Check condition: $V_{GS} > V_{th}$ ($3 > 1$, ON). Check saturation: $V_{GS} - V_{th} = 3 - 1 = 2V$. Since $V_{DS} (4V) > (V_{GS} - V_{th}) (2V)$, the MOSFET is in **Saturation**.

* **Formula:** $I_D = k_n (V_{GS} - V_{th})^2$

* **Calculation:** $I_D = 1 \text{ mA/V}^2 (3 - 1)^2 = 0.5 \text{ mA} \times 4 = 2 \text{ mA}$.

* **Answer:** Saturation Region, $I_D = 2 \text{ mA}$.

LEVEL 3 (NEC Challenge)

Q5: Calculate the small signal transconductance (g_m) and input resistance (r_{π}) of a BJT at room temperature (300K) if it is biased with a collector current of 2.6 mA and has $\beta = 100$.

* **Given:** $I_{CQ} = 2.6 \text{ mA}$, $\beta = 100$, $V_T = 26 \text{ mV}$.

* **Formula:** $g_m = I_C / V_T$, $r_{\pi} = \beta g_m$.

* **Calculation:** $g_m = 2.6 \text{ mA} / 26 \text{ mV} = 0.1 \text{ S} = 100 \text{ mA/V}$.

$r_{\pi} = 100 / 0.1 = 1000\Omega = 1 \text{ k}\Omega$

.

* **Answer:** $g_m = 100 \text{ mA/V}$, $r_{\pi} = 1 \text{ k}\Omega$.

10 NEC-Style MCQs

1. The forbidden energy gap in Silicon at room temperature is approximately:

- A) 0.3 eV
- B) 0.7 eV
- C) 1.1 eV
- D) 5.0 eV

2. In an N-type semiconductor, the Fermi level lies:

- A) exactly in the middle of the forbidden gap.
- B) closer to the conduction band.
- C) closer to the valence band.
- D) inside the valence band.

3. In a reverse-biased PN junction diode, the current is mainly due to:

- A) Majority carriers
- B) Minority carriers
- C) Holes only
- D) Surface leakage only

4. The relation between α and β in a BJT is:

- A) $\beta = \alpha\alpha + 1$
- B) $\beta = \alpha 1 - \alpha$
- C) $\beta = 1 - \alpha\alpha$
- D) $\alpha = \beta - 1\beta$

5. For a BJT to operate in the saturation region:

- A) Both BE and BC junctions must be reverse-biased.
- B) BE junction must be forward-biased and BC reverse-biased.
- C) BE junction must be reverse-biased and BC forward-biased.
- D) Both BE and BC junctions must be forward-biased.

6. Which BJT configuration provides a phase reversal of 180° between input and output?

- A) Common Base
- B) Common Emitter
- C) Common Collector
- D) All of the above

7. In an enhancement-type NMOS, the channel is formed when:

- A) $V_{GS} < V_{th}$
- B) $V_{GS} > V_{th}$
- C) $V_{DS} > V_{GS}$
- D) $V_{DS} = 0$

8. When a MOSFET acts as an amplifier, it is operated in the:

- A) Cut-off region
- B) Linear (Triode) region

- C) Saturation region
- D) Breakdown region

9. The primary advantage of CMOS over other logic families is:

- A) Higher speed
- B) Higher current handling capacity
- C) Extremely low static power consumption
- D) Simpler manufacturing process

10. Small-signal modeling of a BJT is used to analyze its:

- A) DC biasing point
- B) AC response around the Q-point
- C) Maximum power dissipation
- D) Switching speed

Answers to MCQs

1. C | 2. B | 3. B | 4. B | 5. D | 6. B | 7. B | 8. C | 9. C | 10. B

Chapter 1, Section 1.5: Signal Generator

(AExE0105)

Syllabus: Basic Principles of Oscillator, RC, LC and Crystal Oscillators Circuits. Waveform generators.

1. Introduction: The Engineering Problem

In electronics, we often need to create continuous, periodic waveforms (sine waves, square waves, pulses) without any external AC signal source. How can a circuit continuously generate an AC waveform from a pure DC supply?

This is the fundamental problem solved by **oscillators** and **waveform generators**.

An oscillator is a circuit that acts as an energy converter, transforming DC power from the power supply into AC power at a specific frequency and waveform.

Why is this important?

- * **Clock Signals:** Every digital system, microcontroller, and computer requires a precise clock signal to synchronize operations.
 - * **Carrier Waves:** Radio, television, and mobile communications rely on high-frequency sinusoidal carriers for modulation and transmission.
 - * **Test Signals:** Signal generators are essential laboratory tools for testing and characterizing electronic circuits.
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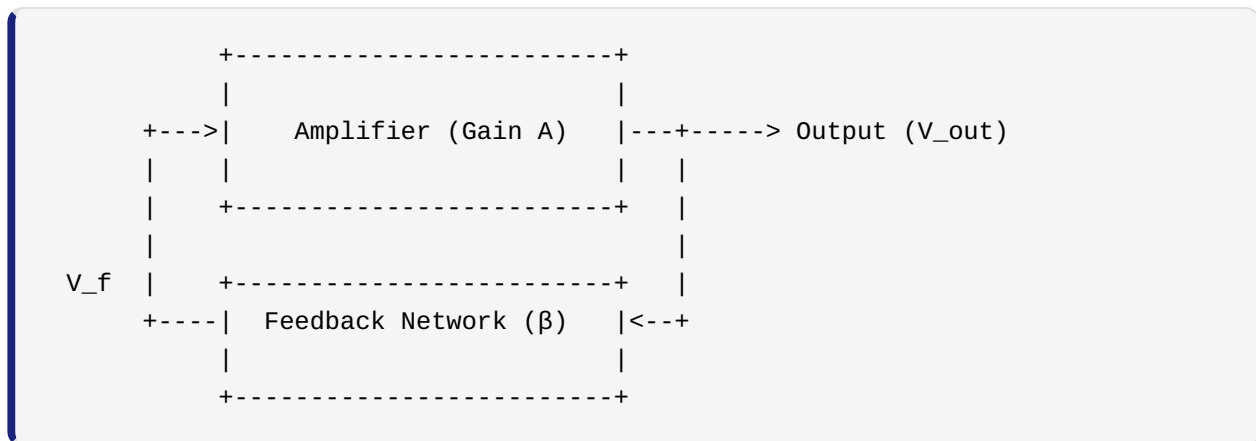
2. Basic Principles of Oscillation

2.1 Amplifier with Feedback

An oscillator is fundamentally an amplifier that provides its own input signal. This is achieved through **positive feedback**.

- **Negative Feedback:** The feedback signal is out of phase (180°) with the input. It opposes the input, leading to stable, predictable amplification but reduced gain.
- **Positive Feedback:** The feedback signal is in phase (0° or 360°) with the original input. It reinforces the input, which can lead to instability and, if controlled, sustained oscillation.

Block Diagram of a Feedback Amplifier:



2.2 Barkhausen Criterion

The Barkhausen Criterion defines the mathematical conditions required for a circuit to produce sustained oscillations.

Statement: For a feedback amplifier to oscillate continuously at a specific frequency, two conditions must be met simultaneously:

1. **Loop Gain Magnitude:** $|A\beta|=1$
2. **Loop Phase Shift:** $\angle A\beta=0^\circ$ (or 360° or $2n\pi$)

Physical Meaning:

Imagine a small noise voltage appearing at the amplifier input. It gets amplified by A , then attenuated and phase-shifted by the feedback network β . For the signal to perfectly recreate itself at the input—neither growing nor shrinking—it must return with the exact

same amplitude ($|A\beta|=1$) and perfectly in phase with itself so it adds constructively ($\angle A\beta=0^\circ$).

Mathematical Derivation:

The closed-loop gain A_f of a feedback amplifier with input V_{in} is:

$$A_f = \frac{V_{out}}{V_{in}} = \frac{A}{1 - A\beta}$$

For an oscillator, there is no external input ($V_{in}=0$), yet there is an output ($V_{out} \neq 0$).

This implies that the closed-loop gain must be infinite ($A_f \rightarrow \infty$).

For A_f to approach infinity, the denominator must be zero:

$$1 - A\beta = 0 \Rightarrow A\beta = 1$$

Since $A\beta$ is a complex quantity, this single equation yields the two Barkhausen conditions for magnitude and phase.

Operating States based on Loop Gain:

- * $|A\beta| > 1$: Growing oscillations. Energy is added to the loop faster than it is dissipated. Necessary for the oscillator to start up from thermal noise.
- * $|A\beta| = 1$: Sustained oscillations (Steady State). Energy added exactly balances energy lost.
- * $|A\beta| < 1$: Decaying oscillations. Energy dissipates, and the oscillation dies out.

2.3 Startup and Amplitude Stabilization

For an oscillator to start, it is designed such that initially $|A\beta| > 1$. Microscopic thermal noise in the circuit provides the initial "kick". Since $|A\beta| > 1$, the amplitude of the chosen frequency grows exponentially.

As the amplitude grows, the amplifier's non-linearities (or an active Automatic Gain Control circuit) cause the gain A to decrease. The amplitude stabilizes exactly at the point where $|A\beta| = 1$.

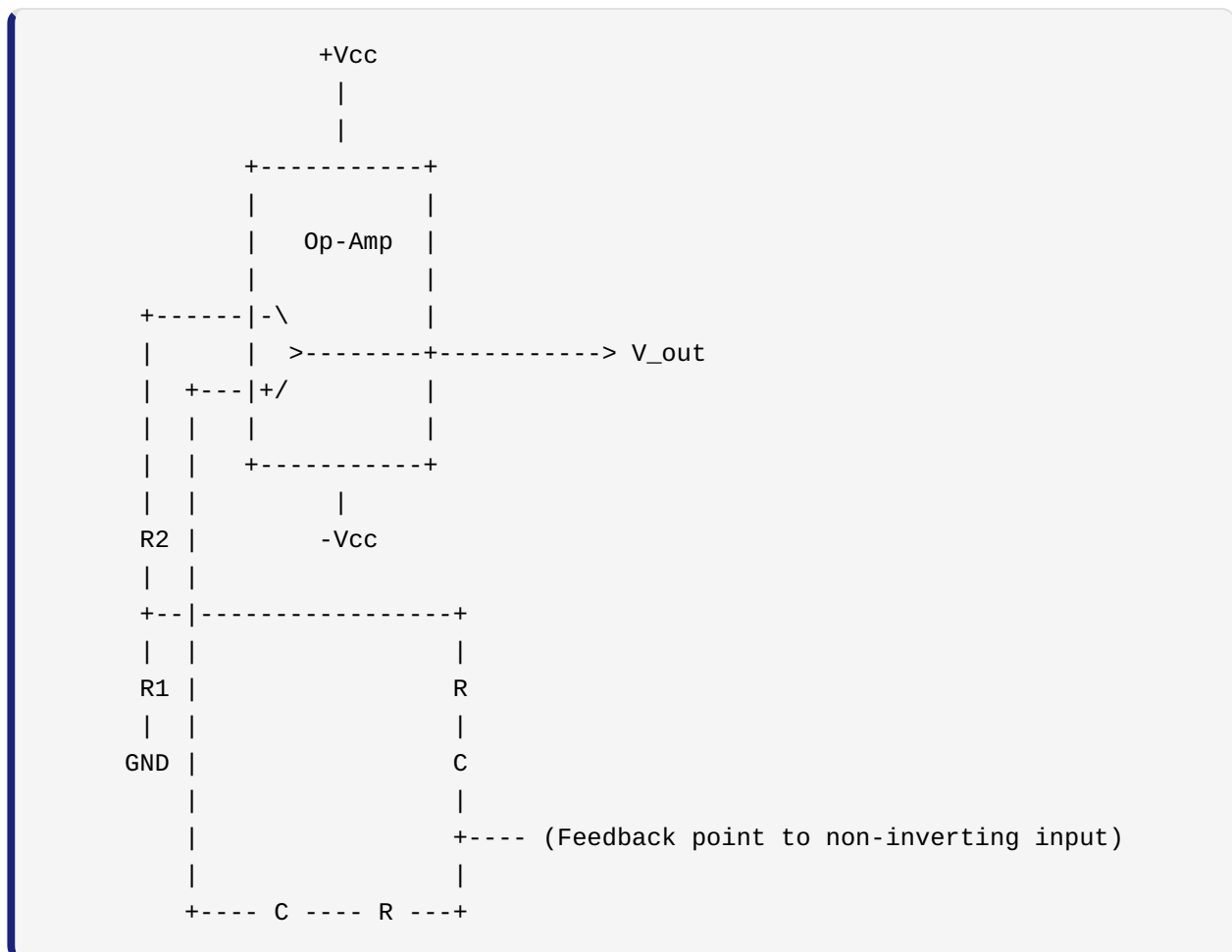
3. RC Oscillators

RC (Resistor-Capacitor) oscillators use RC networks to provide the necessary phase shift. They are typically used for low-frequency oscillations (Audio Frequency, up to around 1 MHz) because LC circuits become impractically large at low frequencies.

3.1 Wien Bridge Oscillator

The Wien bridge oscillator is the standard for generating high-quality, low-distortion sine waves at audio frequencies.

Circuit Diagram (Using Op-Amp):



(Note: R1 and R2 form the negative feedback to set the gain. The series-parallel RC network forms the positive feedback.)

Working Principle:

The positive feedback network consists of a series RC circuit and a parallel RC circuit. At

a specific frequency f_0 , this network provides a phase shift of exactly 0° , and its attenuation factor β is $1/3$.

Frequency of Oscillation Derivation:

Let $Z_1 = R - j1\omega C$ (series arm)

Let $Z_2 = R(-j/\omega C)R - j/\omega C = R1 + j\omega RC$ (parallel arm)

The feedback factor $\beta = V_f/V_{out} = Z_2/Z_1 + Z_2$

Substituting and equating the imaginary part to zero (for 0° phase shift) yields:

$$f_0 = 1/2\pi RC$$

Condition for Oscillation:

At f_0 , the feedback factor $\beta = 1/3$.

To satisfy $|A\beta| \geq 1$, the amplifier gain A must be:

$$A \geq 3$$

Since $A = 1 + R_2/R_1$ for a non-inverting op-amp, this requires $1 + R_2/R_1 \geq 3 \Rightarrow R_2/R_1 \geq 2$.

Characteristics:

* **Advantages:** Excellent frequency stability, very low distortion, wide frequency range easily tuned by varying C .

* **Disadvantages:** Requires two identical variable capacitors or resistors for tuning; limited to audio frequencies.

Worked Numerical Problem (Level 2)

Given: A Wien bridge oscillator uses $R = 10 \text{ k}\Omega$ and $C = 0.01 \text{ }\mu\text{F}$. The feedback resistors are $R_1 = 10 \text{ k}\Omega$ and R_2 .

Required: 1) Frequency of oscillation f_0 . 2) Minimum value of R_2 for sustained oscillations.

Formula: $f_0 = 1/2\pi RC$, $R_2 \geq 2R_1$

Calculation:

1) $f_0 = 1/2\pi \times 10^4 \times 10^{-8} = 1/2\pi \times 10^{-4} \approx 1591.5 \text{ Hz}$

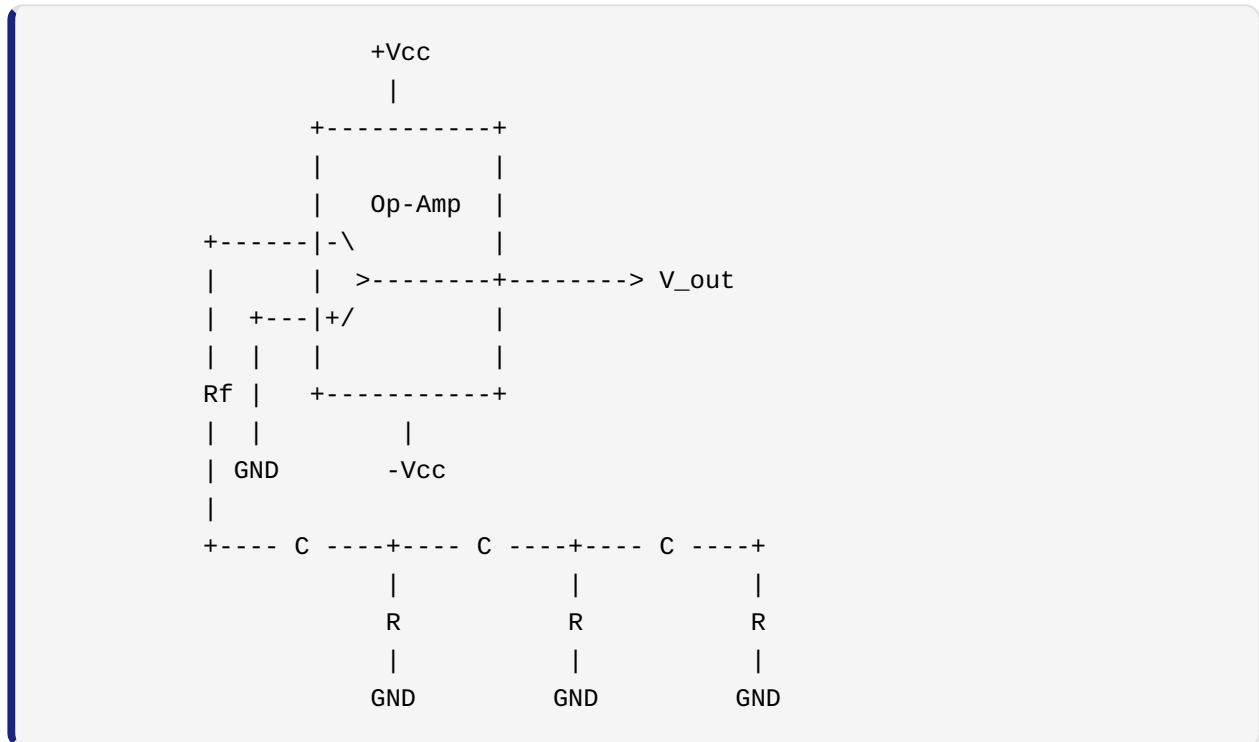
2) For sustained oscillation, $R_2 \geq 2 \times 10 \text{ k}\Omega = 20 \text{ k}\Omega$.

Answer: $f_0 = 1.59 \text{ kHz}$, $R_2 \geq 20 \text{ k}\Omega$.

3.2 Phase-Shift Oscillator

The phase-shift oscillator uses a cascade of RC high-pass filters to achieve the necessary phase shift.

Circuit Diagram (Using Op-Amp):



Working Principle:

The inverting amplifier provides a 180° phase shift. To satisfy the Barkhausen criterion for 0° (or 360°) total phase shift, the feedback network must provide an additional 180° phase shift.

The feedback network uses three RC sections. Each section provides a phase shift of 60° at the resonant frequency ($3 \times 60^\circ = 180^\circ$).

Frequency of Oscillation:

By analyzing the 3-mesh RC network, the frequency where the phase shift is exactly 180° is:

$$f_0 = \frac{1}{12\pi RC}$$

Condition for Oscillation:

At this frequency f_0 , the signal is heavily attenuated by the RC network. The attenuation factor is $\beta = 1/29$.

Therefore, to satisfy $|A\beta| \geq 1$, the inverting amplifier must have a gain magnitude $|A| \geq 29$. For an op-amp, $|A| = R_f/R$, so $R_f \geq 29R$.

Characteristics:

- * **Advantages:** Simple circuit, cheap, good for fixed-frequency audio signals.
- * **Disadvantages:** Difficult to tune (changing one R or C changes the attenuation and frequency unpredictably), requires high gain.

3.3 RC Oscillators Comparison

FEATURE	WIEN BRIDGE OSCILLATOR	PHASE-SHIFT OSCILLATOR
Phase Shift mechanism	Network: 0° , Amp: 0°	Network: 180° , Amp: 180°
Frequency (f_0)	$12\pi RC$	$12\pi RC6$
Minimum Gain ($ A $)	3	29
Tuning	Easy (change C)	Difficult

4. LC Oscillators

LC oscillators use a tuned circuit (an inductor and capacitor in parallel, known as a "tank circuit") to set the frequency. They are used for high frequencies (Radio Frequencies, 100 kHz to hundreds of MHz).

4.1 General LC Oscillator Principle

In an ideal parallel LC circuit, energy sloshes back and forth between the electric field of the capacitor and the magnetic field of the inductor. The natural resonant frequency is:

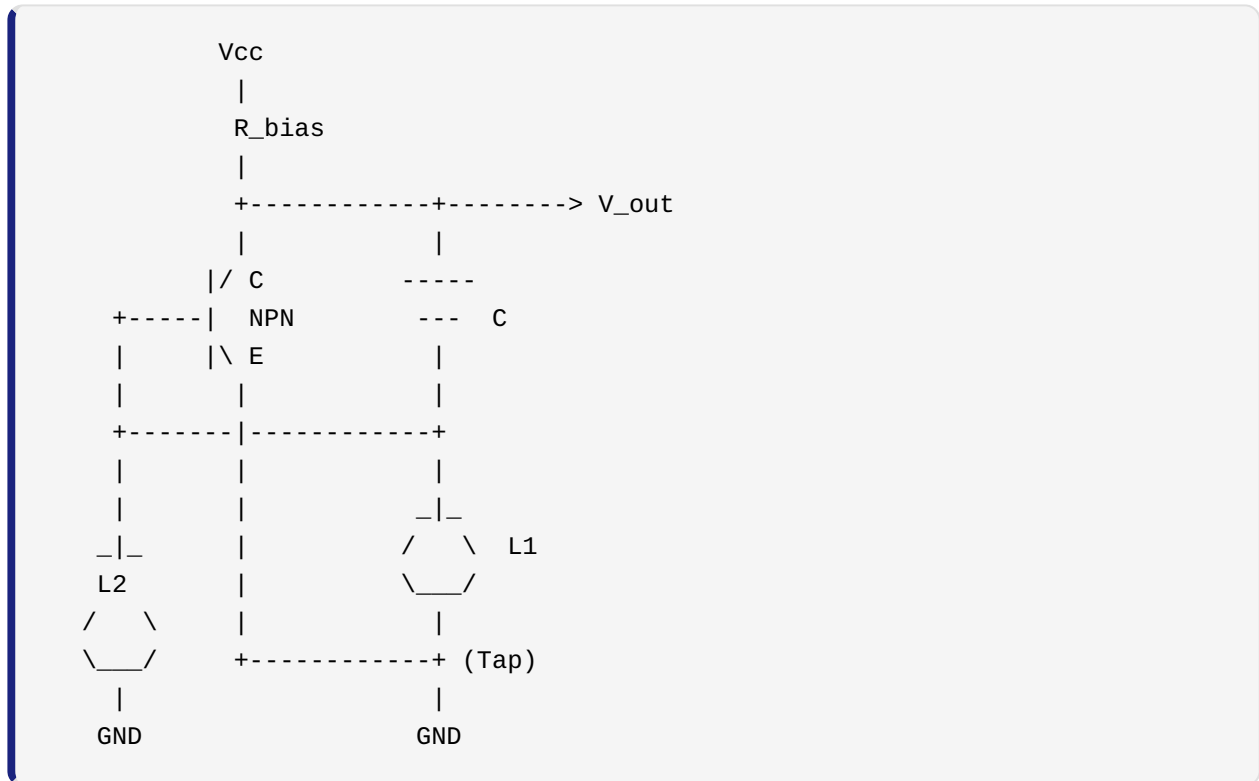
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

However, real inductors have resistance. This causes the oscillations to damp out (decay). The amplifier's job is to inject just enough energy back into the tank circuit to overcome these resistive losses, maintaining sustained oscillations.

4.2 Hartley Oscillator

The Hartley oscillator is distinguished by its tapped inductor.

Circuit Diagram (Conceptual):



Working Principle:

The tank circuit consists of capacitor C in parallel with two series inductors $L1$ and $L2$ (often a single tapped coil). The tap is connected to the common terminal (usually ground in the AC equivalent circuit). The voltage across $L1$ provides the input to the amplifier, and the output is fed across $L2$.

Frequency of Oscillation:

Assuming no mutual inductance ($M=0$):

$$f_0 = \frac{1}{2\pi\sqrt{(L1+L2)C}}$$

If mutual inductance M exists between the coils:

$$f_0 = \frac{1}{2\pi\sqrt{(L1+L2+2M)C}}$$

Condition for Oscillation:

The feedback fraction is determined by the ratio of the inductances. For oscillation:

$$A \geq L_1 L_2$$

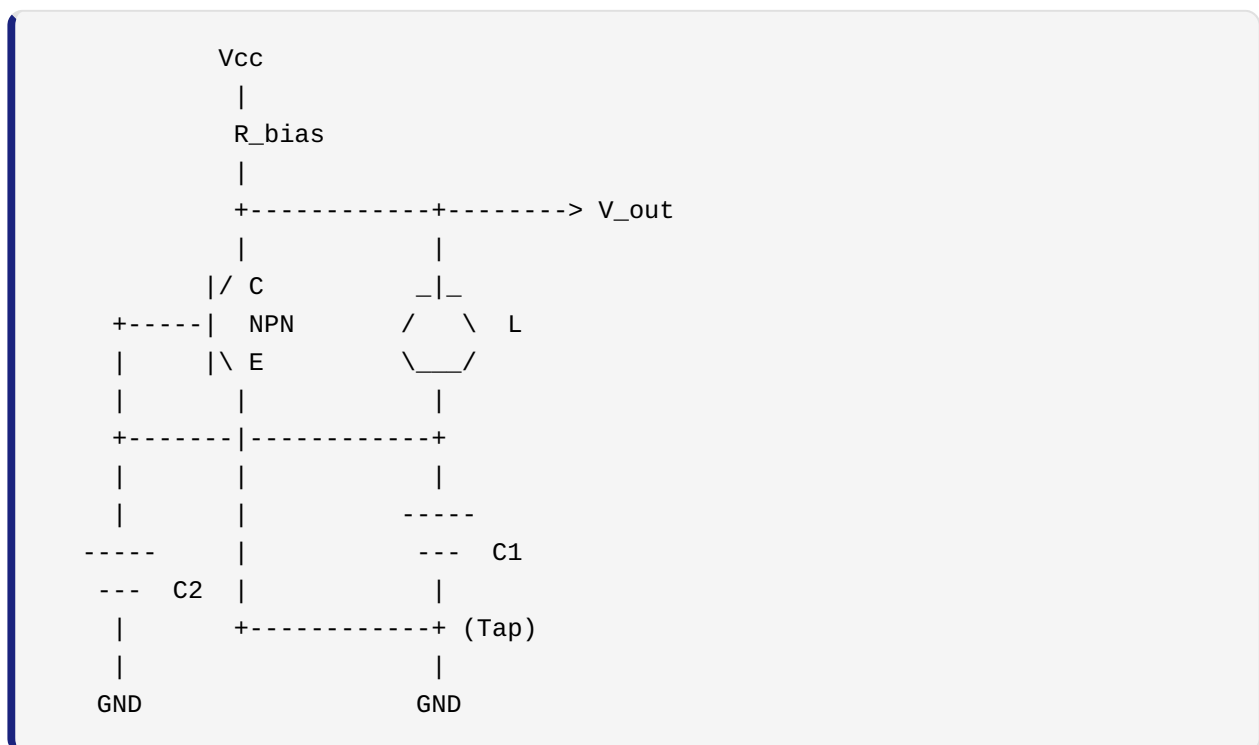
(Note: depending on the exact configuration, the ratio might be L_2/L_1 . The key is that the gain must overcome the step-down ratio of the tapped coil).

Characteristics:

- * **Advantages:** Easy to tune by making C a variable capacitor. Wide frequency range.
- * **Disadvantages:** Harmonic content can be high; poorer frequency stability than Colpitts.

4.3 Colpitts Oscillator

The Colpitts oscillator is the dual of the Hartley oscillator. It uses a tapped capacitor network.

Circuit Diagram (Conceptual):**Working Principle:**

The tank circuit consists of inductor L in parallel with two series capacitors C1 and C2. The junction between C1 and C2 acts as the tap.

Frequency of Oscillation:

The equivalent capacitance C_{eq} of C_1 and C_2 in series is:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

The frequency is:

$$f_0 = \frac{1}{2\pi L C_{eq}}$$

Condition for Oscillation:

The feedback fraction is determined by the capacitor voltage divider:

$$A \geq \frac{C_1 + C_2}{C_2}$$

Characteristics:

- * **Advantages:** Better frequency stability than Hartley. Excellent performance at high frequencies. Lower distortion because the capacitors provide a low-impedance path to ground for high-frequency harmonics.
- * **Disadvantages:** Harder to tune over a wide range since changing both capacitors simultaneously is mechanically complex.

4.4 LC Oscillators Comparison

FEATURE	HARTLEY OSCILLATOR	COLPITTS OSCILLATOR
Tank Circuit	Tapped Inductor (L_1, L_2), Single C	Tapped Capacitor (C_1, C_2), Single L
Feedback Ratio	L_1/L_2	C_1/C_2
Frequency Formula	$\frac{1}{2\pi\sqrt{(L_1+L_2)C}}$	$\frac{1}{2\pi\sqrt{L(C_1 C_2 / (C_1 + C_2))}}$
Best Application	Wideband variable frequency	High-frequency, fixed stability

5. Crystal Oscillators

When extreme frequency stability is required (e.g., in digital watches, computers, transmitters), LC circuits are inadequate because L and C values drift with temperature and aging. Crystal oscillators solve this using the **piezoelectric effect**.

5.1 Piezoelectric Effect

Materials like quartz exhibit piezoelectricity:

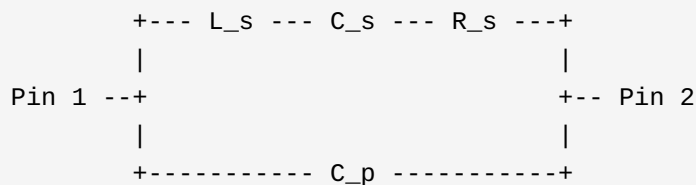
- * Apply mechanical stress → Generates an electrical voltage.
- * Apply an electrical voltage → Crystal mechanically deforms (vibrates).

When an AC voltage is applied, the crystal vibrates. If the AC frequency matches the crystal's mechanical resonant frequency, the vibrations become very strong, and the crystal acts like a highly precise electrical resonant circuit.

5.2 Crystal Equivalent Circuit

Electrically, a quartz crystal behaves like a complex RLC circuit.

ASCII Equivalent Circuit:



- **C_p (Parallel Capacitance):** Capacitance of the mounting electrodes and holder.
- **L_s (Series Inductance):** Represents the crystal's mass.
- **C_s (Series Capacitance):** Represents the crystal's mechanical stiffness (elasticity).
- **R_s (Series Resistance):** Represents internal mechanical friction losses.

Resonant Frequencies:

A crystal has two closely spaced resonant frequencies:

1. **Series Resonant Frequency (f_s):** Due to L_s and C_s. Impedance is minimum.

$$f_s = \frac{1}{2\pi L_s C_s}$$

2. **Parallel Resonant Frequency (f_p):** Due to L_s resonating with the series combination of C_s and C_p . Impedance is maximum.

$$f_p = \frac{1}{2\pi\sqrt{L_s(C_s C_p C_s + C_p)}}$$

Since $C_p \gg C_s$, f_p is slightly greater than f_s (usually by less than 1%). The oscillator usually operates slightly above f_s , where the crystal behaves inductively.

Q Factor:

The quality factor (Q) of a crystal is exceptionally high—ranging from 10,000 to over 1,000,000 (compared to ~ 100 for a good LC circuit). This high Q is the source of its incredible frequency stability.

5.3 Crystal Oscillator Circuits

Crystals can be used to replace the inductor in a Colpitts oscillator (creating a Pierce oscillator) because between f_s and f_p , the crystal exhibits an inductive reactance.

Advantages:

- * Unmatched frequency stability (measured in parts per million, ppm).
- * Extremely high Q factor.
- * Fixed, reliable frequency.

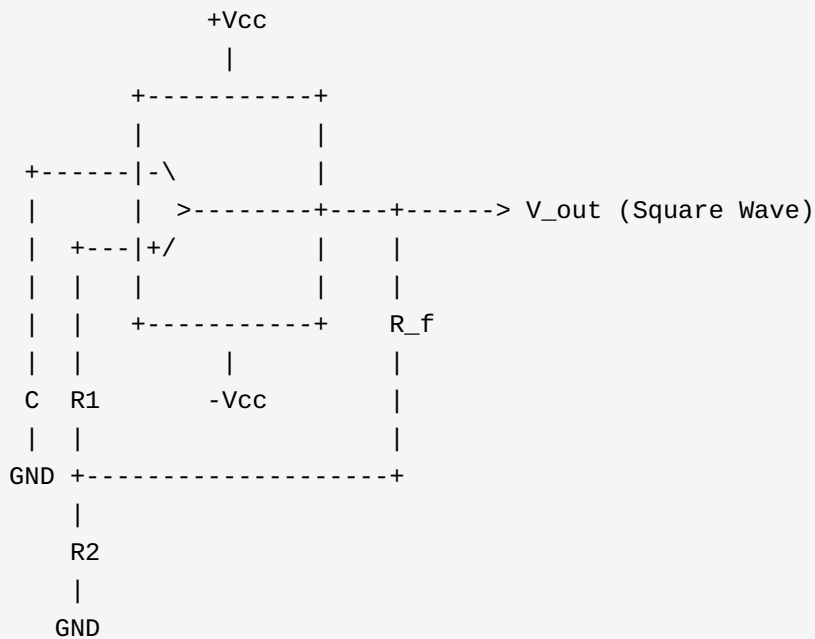
6. Waveform Generators

While harmonic oscillators (RC, LC, Crystal) generate sine waves, many engineering applications require non-sinusoidal waveforms like square, triangular, or sawtooth waves. These are generated using relaxation oscillators or shaping circuits.

6.1 Square Wave Generator (Astable Multivibrator)

An astable multivibrator has no stable states; it continuously switches back and forth between two voltage levels (e.g., $+V_{sat}$ and $-V_{sat}$), producing a square wave.

Circuit Diagram (Op-Amp based):

**Working Principle:**

- * The op-amp acts as a comparator with hysteresis (Schmitt Trigger) due to positive feedback via R1 and R2.
- * Capacitor C charges and discharges through R_f between the upper and lower threshold voltages set by the voltage divider R1, R2.
- * When VC hits a threshold, the output flips to the opposite saturation level, and the capacitor reverses its charging direction.

Frequency Formula:

$$f_0 = \frac{1}{12R_f C \ln(1 + \beta_1 - \beta)}$$

where $\beta = \frac{R_2}{R_1 + R_2}$. If $R_1 = R_2 \times 1.16$, then $f_0 \approx \frac{1}{12R_f C}$.

6.2 Triangular Wave Generator

A triangular wave is created by mathematically integrating a square wave. If you feed a constant DC voltage (the top of a square wave) into an integrator, the output ramps up linearly. When the square wave flips to negative, the integrator ramps down.

Circuit Concept:

[Square Wave Generator] ---> [Op-Amp Integrator] ---> Triangular Wave Out

By combining an astable multivibrator (comparator) and an integrator, we generate both square and triangular waves simultaneously.

6.3 Sawtooth Wave Generator

Similar to a triangular wave, but the rise time is very different from the fall time (e.g., a slow linear ramp up, followed by an almost instantaneous flyback drop).

Applications: Cathode Ray Tube (CRT) sweep circuits, time-base generators in oscilloscopes, PWM modulation.

7. Master Comparison Table

PARAMETER	WIEN BRIDGE	PHASE-SHIFT	HARTLEY	COLPITTS	CRYSTAL
Frequency Range	1 Hz - 1 MHz (AF)	1 Hz - 100 kHz (AF)	100 kHz - 100 MHz (RF)	100 kHz - 100 MHz (RF)	10 kHz - 100 MHz
Feedback Network	Series-Parallel RC	3-stage RC	Tapped Inductor	Tapped Capacitor	Quartz Crystal
Frequency Formula	$12\pi RC$	$12\pi RC6$	$12\pi L_{eq}C$	$12\pi L C_{eq}$	$f_s = 12\pi L_s C_s$
Stability	Good	Poor	Fair	Good	Excellent (Very High)
Q Factor	Low (<10)	Low (<10)	Medium (~100)	Medium (~100)	Extremely High (>10,000)
Applications	Audio Generators	Simple AF sources	Radio Receivers	RF Transmitters	Clocks, Microcontrollers

8. Section Summary & NEC Exam Prep

Key Equations

DESCRIPTION	FORMULA	CONDITIONS
Barkhausen Criterion	$A\beta=1$	Magnitude $ A\beta =1$, Phase $=0^\circ$
Wien Bridge Freq	$f=12\pi RC$	$R1=R2, C1=C2$
Wien Bridge Gain	$A\geq 3$	To overcome $\beta=1/3$
Phase-Shift Freq	$f=12\pi RC6$	3 RC sections
Phase-Shift Gain	$ A \geq 29$	To overcome $\beta=1/29$
Colpitts Freq	$f=12\pi LC_{eq}$	$C_{eq}=C1C2/C1+C2$

⚠ NEC Exam Traps

1. **The 6 Trap:** Forgetting the 6 in the Phase-Shift oscillator formula. $f=12\pi RC$ is ONLY for the Wien bridge.
2. **Gain vs. Feedback:** Barkhausen requires $|A\beta|\geq 1$ to start. If an exam asks for the *feedback factor* β of a Wien bridge, it's $1/3$. The *amplifier gain* required is 3. Do not confuse the two.
3. **Hartley vs. Colpitts Identification:**
 - * **Hartley** = **H**enries (Tapped Inductor).
 - * **Colpitts** = **C**apacitors (Tapped Capacitor).
4. **Crystal Frequencies:** Remember f_p is always slightly greater than f_s .

💡 Engineering Intuition

An oscillator is just an amplifier biting its own tail. The feedback network is a filter that strictly controls *which* frequency is allowed to pass back to the input exactly in phase. Only the frequency that successfully runs the gauntlet of the feedback network without being phase-shifted away from 0° gets continuously amplified.

Conceptual Questions

1. Why is positive feedback required in an oscillator, whereas negative feedback is used in standard amplifiers?
2. Explain the physical significance of the two parts of the Barkhausen criterion.
3. Why are LC oscillators generally not used for very low audio frequencies?
4. What physical property of quartz makes it suitable for use in oscillators?
5. How does a square wave generator fundamentally differ from a harmonic (sine wave) oscillator?

Numerical Problems

Level 1 (Foundation):

1. Calculate the frequency of a Wien Bridge oscillator if $R=5\text{ k}\Omega$ and $C=0.05\text{ }\mu\text{F}$.
2. In a Colpitts oscillator, $L=100\text{ }\mu\text{H}$, $C_1=0.01\text{ }\mu\text{F}$, and $C_2=0.01\text{ }\mu\text{F}$. Find the oscillation frequency.

Level 2 (Engineering):

3. A phase-shift oscillator uses three identical RC sections. If $C=0.1\text{ }\mu\text{F}$ and the desired frequency is 1 kHz, calculate the required value of R and the minimum amplifier gain.

Level 3 (NEC Challenge):

4. An engineer designs a Hartley oscillator with $L_1=2\text{ mH}$, $L_2=1\text{ mH}$, and mutual inductance $M=0.5\text{ mH}$. If the tuning capacitor C is set to 200 pF, calculate the exact oscillation frequency and the minimum required voltage gain of the active device.
5. A quartz crystal has $L_s=0.5\text{ H}$, $C_s=0.05\text{ pF}$, and $C_p=2\text{ pF}$. Calculate both the series resonant frequency and the parallel resonant frequency, and find the percentage difference between them.

NEC-Style MCQs

1. The Barkhausen criterion for sustained oscillations states that the loop gain $|A\beta|$ must be:
 - a) <1
 - b) >1
 - c) $=1$
 - d) $=0$

2. In a Wien bridge oscillator, the minimum voltage gain required for the amplifier to sustain oscillations is:
 - a) 1
 - b) 3
 - c) 29
 - d) 6
3. Which oscillator uses a tapped inductor in its tank circuit?
 - a) Colpitts
 - b) Phase-shift
 - c) Hartley
 - d) Crystal
4. The phase shift provided by the feedback network in a standard 3-stage RC phase-shift oscillator is:
 - a) 0°
 - b) 90°
 - c) 180°
 - d) 360°
5. What is the primary advantage of a crystal oscillator over an LC oscillator?
 - a) Higher output power
 - b) Lower cost
 - c) Exceptional frequency stability
 - d) Wide variable frequency range
6. In a Colpitts oscillator, the feedback fraction is determined by the ratio of:
 - a) Two inductors
 - b) Two capacitors
 - c) Resistor and capacitor
 - d) Mutual inductance to self-inductance
7. A quartz crystal operates on the principle of:
 - a) The Hall effect
 - b) The Piezoelectric effect
 - c) The Seebeck effect
 - d) Electromagnetic induction
8. Which circuit is primarily used to generate a square wave?
 - a) Astable multivibrator
 - b) Integrator

- c) Wien bridge
 - d) Pierce oscillator
9. For an oscillator to start reliably from thermal noise, the initial loop gain $|A\beta|$ should be:
- a) Exactly 1
 - b) Slightly less than 1
 - c) Slightly greater than 1
 - d) Infinity
10. If C_1 and C_2 are the split capacitors in a Colpitts oscillator, the equivalent capacitance C_{eq} determining the frequency is:
- a) $C_1 + C_2$
 - b) $C_1 - C_2$
 - c) $(C_1 C_2) / (C_1 + C_2)$
 - d) $C_1 C_2$
-

Answers to MCQs

- 1. c
 - 2. b
 - 3. c
 - 4. c
 - 5. c
 - 6. b
 - 7. b
 - 8. a
 - 9. c
 - 10. c
-

Section 1.6: Amplifiers

1. Introduction

Amplifiers are fundamental electronic circuits designed to increase the power, voltage, or current of an input signal without significantly distorting its waveform. From boosting a faint microphone signal to driving large loudspeakers, amplifiers form the backbone of modern electronics and communication systems. For the NEC Engineering License Examination, understanding the various classes of output stages, power handling components, tuned circuits, and the ubiquitous Operational Amplifier (Op-Amp) is absolutely critical.

2. Why is it Important?

In real-world engineering, sensors, antennas, and transducers often output signals in the microvolt or millivolt range. These weak signals cannot drive loads like motors, speakers, or transmission lines. Amplifiers bridge this gap, taking small-signal inputs and providing large-signal power outputs. The NEC exam frequently tests the efficiency, biasing, and application of these stages to ensure engineers can design and troubleshoot practical analog systems.

Part A: Amplifier Fundamentals

1. Signal Amplification Concept

Amplification is the process of increasing the magnitude of a variable quantity (voltage, current, or power).

Definition

The **Gain** (A) of an amplifier is the ratio of the output signal to the input signal.

Mathematical Formulation

SYMBOL	MEANING	FORMULA	SI UNIT
A_v	Voltage Gain	V_{out}/V_{in}	Dimensionless (V/V)
A_i	Current Gain	I_{out}/I_{in}	Dimensionless (A/A)
A_p	Power Gain	$P_{out}/P_{in}=A_v \times A_i$	Dimensionless (W/W)

Decibel (dB) Scale:

Because gains can range from 1 to 106, the logarithmic decibel scale is standard in engineering.

- Voltage Gain in dB: $A_v(\text{dB})=20\log_{10}(|V_{out}/V_{in}|)$
- Power Gain in dB: $A_p(\text{dB})=10\log_{10}(P_{out}/P_{in})$



Engineering Intuition: A power gain of +3 dB means the power has doubled. A voltage gain of +20 dB means the voltage has increased by a factor of 10.

Part B: Classification of Output Stages

2. Overview of Output Stages

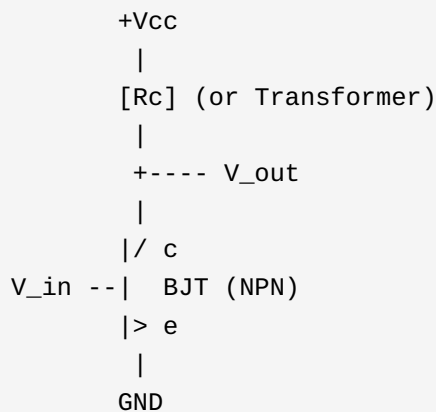
Output stages are classified based on the **conduction angle** (θ), which is the portion of the input sinusoidal cycle (in degrees) during which the amplifying transistor conducts current.

CLASS	CONDUCTION ANGLE	CONDUCTION TIME	EFFICIENCY	DISTORTION
A	$360^\circ (2\pi)$	Full cycle	Low (25%-50%)	Very Low
B	$180^\circ (\pi)$	Half cycle	High (78.5%)	High (Crossover)
AB	$180^\circ < \theta < 360^\circ$	Slightly > half cycle	Medium (50-78.5%)	Low
C	$< 180^\circ$	Less than half cycle	Very High (>90%)	Very High

3. Class A Output Stage

Basic Concept & Circuit

In a Class A amplifier, the transistor is biased such that it remains in the active region for the entire 360° of the input cycle. The Q-point is set exactly in the middle of the AC load line.



Maximum Efficiency Derivation (Resistive Load)

- DC Power Supplied:** $P_{dc} = V_{CC} I_{CQ}$
- For maximum symmetric swing, the Q-point is exactly at $V_{CEQ} = V_{CC}/2$ and $I_{CQ} = I_{C(max)}/2$.

3. Maximum AC Output Power:

$$P_{ac(max)} = V_{p1} I_{p2} = (V_{CC}/2) \times I_{CQ2} = V_{CC} I_{CQ4}$$

4. Efficiency (η):

$$\eta = \frac{P_{ac(max)}}{P_{dc}} \times 100\% = \frac{V_{CC} I_{CQ}}{4 V_{CC} I_{CQ}} = 25\%$$

If a transformer is used instead of a collector resistor, $P_{dc} = V_{CC} I_{CQ}$, but the AC voltage can swing from 0 to $2V_{CC}$, doubling the AC power.

Transformer-coupled Class A efficiency = 50% max.

Engineering Interpretation

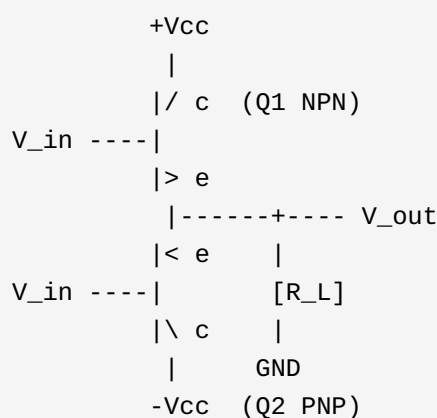
Class A wastes a massive amount of power as heat (especially when there is no input signal), but because the transistor never turns off, it offers the highest linearity and lowest distortion.

- **Applications:** Audio preamplifiers, low-power RF stages.

4. Class B Output Stage

Basic Concept & Circuit

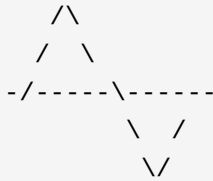
Class B uses a complementary pair of transistors (NPN and PNP) biased exactly at cutoff ($I_{CQ} = 0$). The NPN conducts for the positive half-cycle, and the PNP conducts for the negative half-cycle.



Crossover Distortion

Because BJTs require roughly 0.7V (V_{BE}) to turn on, there is a "dead zone" between $-0.7V$ and $+0.7V$ of the input signal where neither transistor conducts. This causes a flattened spot at the zero-crossing of the output waveform.

Output Waveform with Crossover Distortion:



(Notice the flat step at the axis crossing)

Maximum Efficiency Derivation

1. **AC Output Power:** $P_{ac} = \frac{V_p^2}{2R_L}$ (where maximum $V_p = V_{CC}$)
2. **DC Power Supplied:** Current is half-sine. Average current from one supply = I_p/π .
Total DC power from both supplies = $2 \times V_{CC} \times (I_p/\pi) = \frac{2V_{CC}V_p}{\pi R_L}$
3. **Efficiency:** $\eta = \frac{P_{ac}}{P_{dc}} = \frac{\pi V_p}{4V_{CC}}$. Max efficiency when $V_p = V_{CC}$:

$$\eta_{max} = \frac{\pi}{4} \approx 78.5\%$$

⚠ NEC Exam Trap: Exam questions often ask for efficiency of Class B. Remember 78.5% is the *theoretical maximum*. Actual efficiency is lower depending on signal amplitude.

5. Class AB Output Stage

Class AB solves the crossover distortion of Class B by biasing the transistors *slightly* above cutoff. A small quiescent current flows even with no input signal. The conduction angle is slightly greater than 180° .

- **Efficiency:** Compromise between A and B, practically around 50%-78%.
- **Applications:** Audio power amplifiers (the most common power amplifier class).

6. Biasing the Class AB Stage

To bias the transistors at the edge of conduction, we need a DC voltage drop of $2V_{BE}$ (approx 1.4V) between the bases of the NPN and PNP transistors. This is achieved using:

1. **Diode Biasing:** Two diodes in series.
2. **V_{BE} Multiplier (Rubber Diode):** A transistor with resistors R_1 and R_2 that acts as an adjustable zener diode with voltage $V_{bias} = V_{BE}(1 + R_1/R_2)$.

Thermal Runaway Prevention: By mounting the biasing diodes or V_{BE} multiplier on the same heat sink as the power transistors, their V_{BE} drops decrease with temperature at the same rate, keeping the quiescent current stable.

7. Class A vs Class B vs Class AB Comparison Table

PARAMETER	CLASS A	CLASS B	CLASS AB
Conduction Angle	360°	180°	180° to 360°
Q-Point	Center of active region	Cutoff on load line	Just above cutoff
Max Efficiency	25% (RC) / 50% (Transformer)	78.5%	50% - 78.5%
Distortion	Lowest	Highest (Crossover)	Low
No-Signal Power Dissipation	Maximum	Zero	Very Small
Applications	Preamps	Rarely used alone	Standard audio amplifiers

Part C: Power BJTs

8. Power BJTs

Power BJTs differ from small-signal BJTs in physical size and layout. They are designed to handle tens of amperes and hundreds of volts.

- **Thermal Resistance (θ_{JA}):** The resistance to heat flow from the junction to ambient air. $T_J = T_A + P_D \theta_{JA}$. To keep the junction temperature (T_J) below destruction limits (usually 150°C), **Heat Sinks** are attached to reduce the thermal resistance to ambient.
- **Safe Operating Area (SOA):** A graph of I_C vs V_{CE} showing the boundaries within which the BJT can operate without damage from thermal limits, voltage breakdown, or second breakdown.
- **Second Breakdown:** A localized thermal runaway within the BJT die, causing hot spots and catastrophic failure, even if average power is within limits.
- **Darlington Pair:** Power BJTs have low β (often 20-50). To increase current gain, two BJTs are cascaded. $\beta_{\text{total}} \approx \beta_1 \times \beta_2$.

Part D: Transformer-Coupled Push-Pull Stages

9. Transformer-Coupled Push-Pull Amplifier

```

      +Vcc
      |
      [CT] (Center Tap of Output Transformer)
      |
      | / c
V_in1 ---| (Q1 conducts pos half)
      | > e --- GND
      |
      | / c
V_in2 ---| (Q2 conducts neg half)
      | > e --- GND
  
```

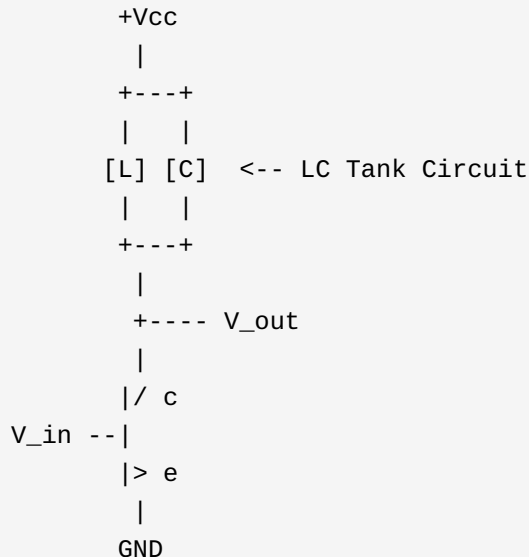
(Input signal is split by a center-tapped input transformer into V_{in1} and V_{in2} , which are 180° out of phase).

- **Working Principle:** One transistor handles the positive half of the signal, the other handles the negative. The output transformer combines them into a full wave.
- **Advantages:** Excellent impedance matching (can drive 8Ω speakers easily), DC isolation.
- **Disadvantages:** Transformers are heavy, bulky, expensive, suffer from core saturation, and have poor frequency response (limits low and high frequencies).

Part E: Tuned Amplifiers

10. Tuned Amplifiers

Tuned amplifiers use an LC (inductor-capacitor) resonant tank circuit as the load instead of a resistor. They amplify only a narrow band of frequencies centered around the resonant frequency.



Physical Meaning

At resonance ($f_0 = 1/2\pi LC$), the impedance of the LC tank is maximum. Since $A_v = -g_m Z_L$, the voltage gain is massive exactly at f_0 and drops off sharply at other frequencies.

Mathematics

- **Resonant Frequency:** $f_0 = \frac{1}{2\pi LC}$
- **Quality Factor (Q):** Ratio of energy stored to energy dissipated. Higher Q = sharper peak.
- **Bandwidth (BW):** $BW = f_0/Q$

Applications

Radio receivers (selecting one station out of many), IF amplifiers, RF transmitters.

Part F: Operational Amplifiers (Op-Amps)

11. Ideal vs Practical Op-Amp (12 & 13)

PARAMETER	IDEAL OP-AMP	PRACTICAL OP-AMP (E.G., 741)
Voltage Gain (AOL)	Infinite (∞)	105 to 106
Input Impedance (Z_{in})	Infinite (∞)	$\approx 2 \text{ M}\Omega$
Output Impedance (Z_{out})	Zero (0Ω)	50–75 Ω
Bandwidth	Infinite	Finite (Gain-Bandwidth Product limited)
CMRR	Infinite	>90 dB
Input Currents (I_+, I_-)	Zero	Nanoamps (Bias current)

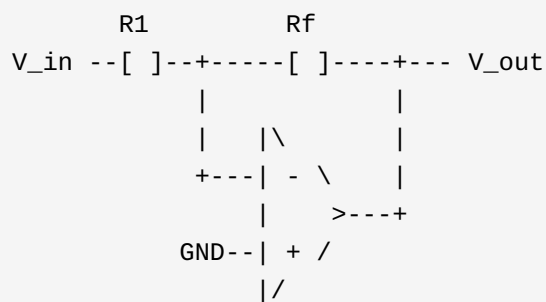
14. Key Concepts for Solving Op-Amp Circuits

When an op-amp has **Negative Feedback** (connection from output to inverting (-) input), it behaves according to two golden rules:

1. **Virtual Short:** The op-amp drives the output to whatever voltage is necessary to make the voltage difference between the inputs zero. Therefore, $V_+ \approx V_-$.
2. **Zero Input Current:** $I_+ = I_- = 0$.

(If V_+ is grounded, V_- becomes a "Virtual Ground", meaning it is at 0V but can accept current through feedback components).

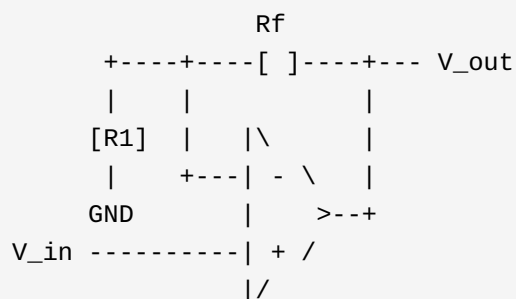
15. Inverting Amplifier



Derivation:

1. Node at inverting input is virtual ground ($V_- = 0V$).
 2. KCL at V_- : $I_{in} + I_f = 0 \Rightarrow V_{in} - 0R_1 + V_{out} - 0R_f = 0$
 3. Gain: $A_v = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$
- **Input Impedance:** $Z_{in} = R_1$.

16. Non-Inverting Amplifier



Derivation:

1. Virtual short: $V_- = V_+ = V_{in}$.
2. KCL at V_- : $0 - V_{in}R_1 + V_{out} - V_{in}R_f = 0$
3. Gain: $A_v = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_1}$

17. Voltage Follower (Unity Gain Buffer)

Connect output directly to inverting input ($R_f=0, R_1=\infty$).

- Gain = 1 ($V_{out}=V_{in}$).
 - **Application:** Buffer to prevent loading effects (high input impedance, low output impedance).
-

18. Summing Amplifier (Inverting)

Uses multiple input resistors connected to the virtual ground node.

- **Formula:** $V_{out} = -R_f(V_1/R_1 + V_2/R_2 + \dots)$
 - **Application:** Audio mixing consoles, Digital-to-Analog Converters (DAC).
-

19. Difference Amplifier

Amplify the difference between two inputs.

- **Formula:** If $R_1=R_2$ and $R_f=R_3$, then $V_{out}=R_f R_1(V_2-V_1)$
 - **Application:** Instrumentation, bridge sensor outputs.
-

20. Integrator

Replace R_f with a Capacitor C .

- **Formula:** $V_{out} = -1/RC \int V_{in} dt$
- **Application:** Converts square wave to triangle wave; analog computing.

21. Differentiator

Capacitor at input, Resistor at feedback.

- **Formula:** $V_{out} = -RC dV_{in}/dt$
- **Application:** Waveform edge detection. *Practical note:* Highly susceptible to high-frequency noise.

22. Comparator

No negative feedback (Open loop).

- If $V_+ > V_-$, $V_{out} = +V_{sat}$ (approx $+V_{CC}$)
- If $V_+ < V_-$, $V_{out} = -V_{sat}$ (approx $-V_{CC}$)
- **Application:** Analog-to-digital conversion, zero-crossing detectors.

End-of-Section Content

Key Equations

CONCEPT	EQUATION
Class A Max Efficiency	$\eta = 25\%$ (RC)
Class B Max Efficiency	$\eta = \pi/4 \approx 78.5\%$
Tuned Amp Bandwidth	$BW = f_0/Q$
Inverting Op-Amp Gain	$A_v = -R_f/R_1$
Non-Inverting Op-Amp Gain	$A_v = 1 + (R_f/R_1)$

5 Conceptual Questions

1. Why is Class A amplification not used in high-power battery-operated devices?
2. Explain the physical origin of crossover distortion in a Class B amplifier.
3. How does a VBE multiplier help stabilize a Class AB amplifier against temperature changes?
4. Why is the input impedance of an inverting amplifier practically equal to R_1 , whereas the non-inverting amplifier has nearly infinite input impedance?
5. Why are practical differentiator circuits rarely used without an input resistor in series with the capacitor?

5 Numerical Problems

Level 1 (Foundation)

1. Calculate the voltage gain in dB of an amplifier with $V_{in}=10\text{ mV}$ and $V_{out}=2\text{ V}$.
2. A non-inverting op-amp circuit uses $R_f=47\text{ k}\Omega$ and $R_1=1\text{ k}\Omega$. Find its voltage gain.

Level 2 (Engineering)

3. A Class A transformer-coupled amplifier drives an 8Ω speaker. If $V_{CC}=12\text{V}$ and the quiescent current is 1A , calculate the DC power consumed, and the maximum possible AC power delivered to the speaker.
4. Design an inverting summing amplifier to output $V_{out}=-(2V_1+5V_2)$. Select a feedback resistor $R_f=10\text{ k}\Omega$ and calculate R_1 and R_2 .

Level 3 (NEC Challenge)

5. **Given:** A tuned amplifier with an LC tank where $L=1\text{mH}$ and $C=1\text{nF}$. The tank has a quality factor $Q=50$.

Required: Resonant frequency and Bandwidth.

Calculation: $f_0=1/(2\pi\sqrt{LC})\approx 159.15\text{ kHz}$. $BW=f_0/Q=159.15/50=3.18\text{ kHz}$.

Interpretation: This amp will only amplify signals strongly between roughly 157.5 kHz and 160.7 kHz .

10 NEC-Style MCQs

1. The maximum theoretical efficiency of a Class B push-pull amplifier is:
A) 25%
B) 50%
C) 78.5%
D) 100%
2. Crossover distortion is a major disadvantage of which amplifier class?
A) Class A
B) Class B
C) Class AB
D) Class C
3. In a Class AB amplifier, the conduction angle is:
A) Exactly 180°
B) Exactly 360°
C) Less than 180°
D) Between 180° and 360°

4. The purpose of a heat sink in power transistors is to decrease the:
A) Thermal resistance from junction to ambient
B) Safe Operating Area
C) Base current
D) Current gain
5. In a tuned amplifier, the bandwidth is given by the ratio of:
A) Resonant frequency to inductance
B) Inductance to Quality Factor (Q)
C) Resonant frequency to Quality Factor (Q)
D) Quality factor to Resonant frequency
6. An ideal operational amplifier has an open-loop voltage gain of:
A) Zero
B) 1
C) 100,000
D) Infinite
7. The virtual ground concept in an op-amp relies on:
A) Infinite input bias current
B) Positive feedback
C) Infinite open-loop gain and negative feedback
D) Zero output impedance
8. For an inverting amplifier with $R_1=2\text{k}\Omega$ and $R_f=10\text{k}\Omega$, the voltage gain is:
A) 5
B) -5
C) 6
D) -0.2
9. A voltage follower is constructed using an op-amp. Its primary application is:
A) Voltage amplification
B) High-frequency oscillation
C) Impedance matching (buffer)
D) Integration of signals
10. If an op-amp circuit replaces the feedback resistor with a capacitor, it becomes a/an:
A) Integrator
B) Differentiator
C) Oscillator
D) Summing amplifier

Answers to MCQs

1. C, 2. B, 3. D, 4. A, 5. C, 6. D, 7. C, 8. B, 9. C, 10. A
